



EEG-based motor imagery classification with quantum algorithms

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ABSTRACT

Developing efficient algorithms harnessing the power of current quantum processors has sparked the emergence of techniques that combine soft computing with quantum computing. This paper proposes two methods that effectively exploit quantum processors for electroencephalography-based motor imagery classification. The first method is a wrapper feature selection approach based on a quantum genetic algorithm, while the second approach employs a hybrid classical-quantum network comprising a baseline feature extraction network and a variational quantum circuit serving as the classifier. Our evaluation on the BCI Competition IV dataset 2b yielded competitive mean accuracies of 83.82%, 85.56%, and 73.73% for the subject-dependent cross-validation, subject-dependent hold-out validation, and subject-independent leaving one subject out classification approaches, respectively. Notably, our statistical analysis revealed that the hybrid models performed on par with the majority of state-of-the-art architectures, underscoring the practical viability of quantum-hybrid methodologies in real-world problem-solving scenarios.

1. Introduction

Since its conception (Feynman & Shor, 1982), quantum computing has been seen as a revolutionary technology with the potential to solve complex problems that are currently beyond the reach of classical computers. Nevertheless, the quest for quantum speedups is subject to the availability of a fault-tolerant quantum computer that is still years and thousand qubits away from reality. We are currently in the so-called NISQ era (Preskill, 2018), where medium-scale devices have emerged as special-purpose processing units for specific tasks. However, the inherent noise in these systems introduces errors into quantum computations, thereby restricting the scale of quantum circuits that can be reliably used. This limitation poses a barrier to the potential accomplishments of quantum devices in the near future.

A popular prospect application for quantum computing in the NISQ era is machine learning. Quantum machine learning (QML) is a rapidly developing field that uses quantum computing design to study traditional machine learning (ML) problems (Schuld & Killoran, 2022). The goal of QML is to leverage the power of quantum computing to develop more efficient and effective ML algorithms. Despite the promising benefits shown by some proof-of-principle studies under certain circumstances (Hayashi & Yang, 2023; Huang et al., 2022; Liu et al., 2021; Watkins et al., 2023), the practical quantum advantage in ML applications remains a significant challenge. Although current

ML algorithms are powerful in practice they are difficult to analyze theoretically, particularly in deep learning. Quantum computing, on the other hand, does not provide useful benchmarks on realistic scales and its applicability for a particular problem is primarily evaluated through theoretical analysis.

The high expectations surrounding quantum computing have led the QML research main focus to be demonstrating quantum advantage; often relying on biased data sets that favor quantum models and exploring abstract scenarios that are not practical from an industrial standpoint (Abbas et al., 2021; He et al., 2023; Huang et al., 2021; Park et al., 2023). Therefore, concentrating solely on quantum advantages may restrict and bias the scope of research and prevent the exploration of new and innovative use cases for quantum computing in ML (Schuld & Killoran, 2022).

The current focus of QML algorithm development is to maximize the potential of NISQ devices by employing parameterized quantum circuits (PQCs) (Cerezo et al., 2021), which operate similarly to neural networks. In practice, the term quantum neural network (QNN) refers to PQCs (Das & Bolisetti, 2023). These circuits have fixed-size structures with adjustable parameters that can be optimized through classical computation to perform specific tasks. In contrast with traditional quantum algorithms, variational quantum algorithms (VQAs) require fewer qubits and less quantum operation depth on the quantum

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processor, making them promising NISQ algorithms (Farhi & Neven, 2018).

According to a recent survey conducted by Melnikov et al. (2023), quantum neural networks, hybrid quantum–classical algorithms, and industrial applications are among the present and near future trending topics in QML. Hybrid quantum–classical ML algorithms have been successfully applied to image classification tasks with the aim of studying if these algorithms are capable of having comparable performance to classical algorithms. Over the past five years there has been a growing interest in the utilization of quantum computing and QML algorithms in medicine (Flöther, 2023). Specifically, there have been notable advancements in genomics research, diagnostics, and the development of new treatments (Flöther, 2023; Wei et al., 2023).

Taking into account these trends and the current state-of-the-art in QML a challenging application is the processing of data with complex structures such as electroencephalographic (EEG) signals (Flöther, 2023). This paper introduces two approaches for classifying motor imagery (MI) tasks based on EEG using quantum algorithms that can be executed on NISQ devices. The first approach is a wrapper feature selection method based on a quantum genetic algorithm which utilizes hand-engineered features and a conventional machine learning classifier. The second approach relies on an end-to-end hybrid classical–quantum network that inputs raw data, automatically extracts features using a baseline convolutional neural network, and performs QNN classification.

This paper presents three main contributions:

- We introduced a novel quantum genetic algorithm specifically designed for feature selection on NISQ devices. The key features of this algorithm include the low quantum depth required to perform population evolution along with its tailored design which optimizes efficient execution on NISQ devices. This algorithm effectively addresses the challenge of dimensionality reduction in MI classification showcasing the potential of quantum methods in enhancing feature selection processes.
- We delved into the potential of both quantum and hybrid classical–quantum algorithms for EEG-based MI classification—an area domain that has received limited attention in the existing literature. Despite the inherent similarities between the functioning of the human brain and quantum computers, few studies have explored their combination to improve classification accuracy. The primary goal of this exploratory study was not to outperform existing methods but to deliver statistically comparable results when running part of the model on a NISQ device. In this context, the proposed hybrid models demonstrated competitive performance when compared to the state-of-the-art, even when employing generic feature extraction networks and a general-purpose ansatz. Our investigation sheds light on the challenges and untapped benefits of merging classical and quantum approaches in this domain, opening new avenues for enhanced classification accuracy.
- Lastly, we demonstrated the practical feasibility of NISQ algorithms in tackling real-world problems of great importance. Compared to many state-of-the-art QML proposals limited to simulations or toy examples, our work explores the application of quantum algorithms to practical problems, providing tangible evidence of their effectiveness. The overarching purpose of this work is to advance the field of quantum machine learning by enhancing our understanding of how quantum techniques can be effectively harnessed, as well as their capabilities and limitations in addressing real-world problems. The main research focus is then on pursuing knowledge and insights rather than exclusively seeking quantum advantages.

The composition of the paper is as follows: In Section 2, we discuss the state-of-the-art techniques for EEG-based MI and QML techniques; in Section 3, we describe the first method based on a quantum genetic

algorithm for feature selection; in Section 4, we describe the second proposed method based on a hybrid classical–quantum neural network; in Section 5, we describe the experiments and discuss their results; finally, in Section 6 we present the conclusions.

2. Background

The following subsections describe state-of-the-art proposals for EEG-based MI and quantum machine learning applied to EEG-based MI classification. Additionally, we describe the theoretical foundations of quantum computing used for our method.

2.1. Related works

Brain–computer interface (BCI) is a communication system that translates brain signals into commands or instructions to control an external device or interact with the environment directly (Wolpaw et al., 2002). Electroencephalography is a noninvasive brain data acquisition method widely used in BCI research due to its low cost, high temporal resolution, and portability (Khosla et al., 2020). EEG measures and records the electrical activity of the brain using multiple electrodes placed on specific scalp areas with millisecond functional accuracy (Amin et al., 2019).

One of the most known and used EEG-based BCI paradigms is motor imagery (MI), which is based on the evidence that the imagination of motor tasks activates similar brain pathways as the real execution of movements (Alazrai et al., 2019; Jackson et al., 2001). A motor imagery brain–computer interface (MI-BCI) is a system that converts the brain signals generated during such imagination into external commands without actual body movements (Li, Li et al., 2023; Tibrewal et al., 2022).

Traditional EEG-based MI classification uses feature extraction techniques and machine learning methods, which are well adapted to the small size of EEG datasets. The most widely used feature extraction algorithms for this purpose are common spatial pattern (CSP) (Müller-Gerking et al., 1999) and its variants in the space-domain (Ang et al., 2012; Lin & Chen, 2023; Zhang, Zhou et al., 2015), and wavelet transform (WT) approaches in the time–frequency domain (Taran & Bajaj, 2019; You et al., 2020). Similarly, popular conventional algorithms for feature classification of MI tasks include linear discriminant analysis (LDA) (Guo et al., 2020; Hsu, 2011), Bayesian classifiers (Wang et al., 2020), support vector machines (SVM) (Hou et al., 2022; Mebarkia & Reffad, 2019), and k-nearest neighbors (KNN) (Chaudhary et al., 2020).

In EEG-based MI-BCI systems, the high dimensionality of the data can often hinder the classifier's performance. This issue is addressed using a feature selection step, commonly applied after feature extraction, to remove noisy, redundant, or irrelevant features and enhance the classifier's performance. The wrapper method is a widely used feature selection technique that employs a predictor and its output as an objective function to evaluate the subset of features. Although wrapper methods tend to be computationally expensive, they often achieve higher accuracy than filter techniques (Xue et al., 2016).

Regarding the use of Quantum-inspired metaheuristics (QIMs) for MI BCI, Kirar and Agrawal (2020) propose a two-phase hybrid model that merges feature selection based on statistical features and a classification algorithm that selects the feature subset. This method finds the relevant and non-redundant electrodes for effective MI classification. The first stage of the model uses a spectral graph to rank electrodes and find a reduced subset of electrodes. The second step uses a quantum genetic algorithm to further reduce the number of electrodes and find the subset that provides the highest classification accuracy via an SVM classifier.

In a recent proposal (Li et al., 2022), a quantum-inspired methodology is also developed for EEG signals. First, the authors prepare the EEG signal as a quantum state preserving the sign of the input data. Then, the prepared quantum state is processed with a quantum wavelet

packet transformation, and a wavelet packet energy entropy feature is extracted as the input of a quantum classifier. The classifier is a quantum support vector machine with an arbitrary nonlinear kernel.

In motor imagery, only a handful of studies have utilized QIM for feature selection, including the works by [Ramos and Vellasco \(2018, 2020\)](#). These studies proposed using a quantum-inspired evolutionary algorithm (QIEA) and a chaotic quantum-inspired evolutionary algorithm as wrapper methods for feature selection in EEG-based MI classification. Nevertheless, these algorithms cannot be directly applied to quantum devices as they employ classical operators for faster convergence.

Deep learning techniques have recently gained attention in the MI classification field because they can learn distinct and latent features directly from raw EEG data without preprocessing or manual feature extraction ([Altaheri et al., 2023](#)). Different DL architectures have been proposed for MI classification, with convolutional neural networks (CNN) and recurrent neural networks (RNN) being the most popular ([Li & Sun, 2022; Liu et al., 2023](#)).

Several CNN models have been suggested within the field of end-to-end architectures, such as Deep ConvNet ([Schirrmester et al., 2017](#)), and EEGNet ([Lawhern et al., 2018](#)), that have become the standard for EEG signal decoding. Deep ConvNet employed a temporal and spatial convolutional layers structure designed to extract band-power features, with network depths and filter sizes that vary between the two models. EEGNet is a compact CNN based on depthwise and separable convolutions to classify EEG signals across various BCI paradigms. Despite their success in EEG decoding at the time, the utilization of 2D convolutions with a single-scale convolution kernel posed limitations in effectively addressing individual differences and exploring spatial relationships, leading to constrained classification accuracy.

Numerous CNN variants have been proposed to overcome these limitations by building upon the architectures of the methods mentioned above, including multi-scale CNN models. HS-CNN ([Dai et al., 2020](#)) is a hybrid-scale CNN with a novel data augmentation method that leverages information from multiple frequency bands of raw EEG signals as inputs to improve classification accuracy. EEG-Inception ([Zhang et al., 2021](#)) has a backbone consisting of six inception modules and two residual modules. The model incorporated multiple 1-D convolutional layers designed to extract parallel features from different time lengths.

To fully extract the temporal and spatial features of EEG signals, [Tang et al. \(2023\)](#) proposed a novel hybrid network: MSHCNN. This network combined multi-scale 1D convolutions to extract advanced temporal features and 2D convolutions to extract temporal and spatial features. In addition, the authors introduced a robust channel coding method to enhance the features of EEG signals. The model obtained a high accuracy in intra-subject, cross-subject, and online experiments. The MTFB-CNN, introduced by [Li, Chen et al. \(2023\)](#), is a novel end-to-end model with superior performance in decoding EEG signals. The MFTB-CNN can effectively extract the time, frequency, and time–frequency features from raw EEG data through parallel multi-scale time–frequency blocks and then performs feature selection by fusing and filtering the features through an attention mechanism and a residual module.

Although recent studies have shown promising results in decoding MI-EEG signals, several challenges remain, as highlighted in the review conducted by [Khademi et al. \(2023\)](#). These challenges include needing separate model training for each subject due to individual differences in MI-EEG signals, resulting in high computational costs. Additionally, training deep learning models on limited MI-BCI datasets is challenging due to the large number of parameters involved, often requiring a significant amount of training data to mitigate overfitting.

QML methods hold significant promise for medical diagnosis, primarily due to their ability to decompose and comprehend high-dimensional and complex data structures using mechanisms beyond the capabilities of classical methods ([Flöther, 2023](#)). [Bowles et al. \(2023\)](#) suggested that identifying data structures that are naturally

suited for quantum models and difficult to emulate classically can provide valuable insights into the potential of quantum machine learning. Additionally, recent findings implying the similarities between the functioning of the human brain and quantum computers ([Chawla & Morgera, 2022; Gao, 2023](#)) have sparked interest in applying quantum techniques to the analysis of EEG signals.

In the context of EEG data analysis using QML, [Gandhi et al. \(2015\)](#) proposed using a Recurrent Quantum Neural Network based on the Schrödinger wave equation to filter MI EEG signals. The authors compared the performance of the RQNN against the Savitzky-Golay filtering model using BCI competition IV dataset 2b. The results indicate that the RQNN improves the accuracy of the training and evaluation sets. In other work, [Aishwarya et al. \(2020\)](#) tested different QML techniques, such as a hybrid quantum convolutional neural network, a variational quantum classifier, and a quantum annealing classifier to detect if consumers liked or disliked a product with the Neuromarketing dataset.

By applying QML algorithms to diverse problem domains, we can enhance our understanding of quantum computing, explore its applicability in solving various problems and uncover the algorithms that excel in specific contexts. However, since scaling a purely quantum algorithm is complicated due to limitations in quantum resources, hybrid classical–quantum approaches alleviate the need for higher-complexity quantum processing by implementing some subroutines on classical hardware. In the context of fully quantum designs, [Farhi and Neven \(2018\)](#) proposed a PQC design for handwritten digits recognition, while [Das and Bolisetti \(2023\)](#) presented a PQC for tasks involving the MNIST and Crack detection datasets.

Conversely, several recent approaches have incorporated dimensionality reduction followed by the application of variational quantum circuits. For instance, [Shi et al. \(2023\)](#) harnessed a PCA in conjunction with a PQC. There is also a noteworthy trend focusing on the fusion of deep learning layers as encoders, combined with PQCs for image classification. The aforementioned approach has been adopted in the works of [Alam et al. \(2021\)](#) and [Sagingalieva et al. \(2023\)](#). In both studies, deep neural networks were used to encode information and subsequently relay high-level features to a PQC.

Our proposal aligns closely with the approach taken by [Alam et al. \(2021\)](#), given the utilization of deep-layer-based dimensionality reduction and encoding techniques. However, we extend their framework to address spatiotemporal problems, incorporating more extensive fine-tuning and in-depth analysis of quantum algorithms.

2.2. Quantum computing fundamentals

Quantum computers derive power from their unique approach to representing and manipulating information. Instead of traditional bits, quantum computers utilize special information units known as quantum bits or qubits that describe a two-dimensional quantum system. A generic qubit in its pure state can be geometrically represented as:

$$|\psi\rangle = \exp(i\gamma) \left(\cos \frac{\theta}{2} |0\rangle + \exp(i\phi) \sin \frac{\theta}{2} |1\rangle \right), \quad (1)$$

where θ , ϕ and γ are real numbers, and $0 \leq \theta \leq \pi$ and $0 \leq \phi \leq 2\pi$. Since the global phase factor $\exp(i\gamma)$ has no observable effect, only the polar θ and azimuth ϕ angles are considered.

A set of n qubits is known as a quantum register. When these qubits are not entangled, the system can be described by a tensor product of single qubits q of the form: $|\psi\rangle = |q_1\rangle \otimes \dots \otimes |q_n\rangle$. This representation is referred to as separable. However, when the qubits are entangled, the state $|\psi\rangle$ is no longer separable, and the system can be expressed as:

$$|\psi\rangle = \alpha_0 |0 \dots 00\rangle + \alpha_1 |0 \dots 01\rangle + \dots + \alpha_{2^n-1} |1 \dots 11\rangle = \sum_{k=0}^{2^n-1} \alpha_k |k\rangle, \quad (2)$$

with $\alpha_k \in \mathbb{C}$ and $\sum_{k=0}^{2^n-1} |\alpha_k|^2 = 1$ ([Williams, 2011](#)). With Eq. (2) we can create a computational basis for an n qubit system as $\{|0 \dots 00\rangle, |0 \dots 01\rangle, \dots, |1 \dots 11\rangle\}$.

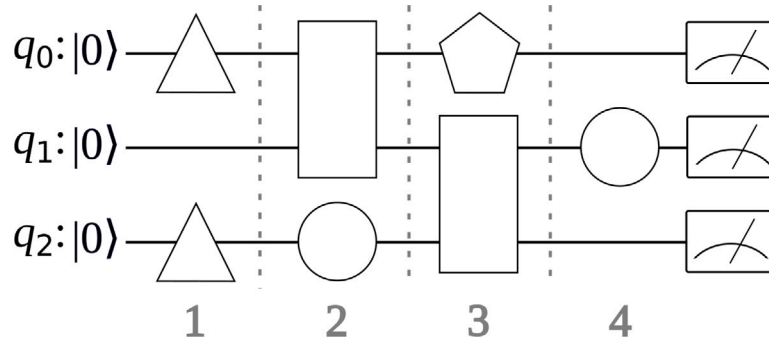


Fig. 1. Illustration of a quantum circuit.

Information from quantum registers are attained by performing projective measurements over the qubits. A projective measurement can be defined by an observable $\mathcal{O}$, which is a Hermitian operator acting on the state space of the system being measured. This operator has a spectral decomposition $\mathcal{O} = \sum_k \alpha_k P_k$, where P_k is the projector onto the eigenspace of $\mathcal{O}$ with eigenvalue α_k . Due to the Hermitian property, the eigenvalues α_k of the observable are guaranteed to be real and constitute the possible outcomes of the measurement (Nielsen & Chuang, 2010).

After a quantum system is measured, the wave function describing the state collapses along with the superposition of possible states into a single definite outcome. The probability of observing an eigenvalue α_k is given by $p(\alpha_k) = \langle \psi | P_k | \psi \rangle$. Given that the outcome k occurred, the post-measurement state of the quantum system is given by:

$$\frac{P_k |\psi\rangle}{\sqrt{p(\alpha_k)}}. \quad (3)$$

In contrast with classical measurements, quantum measurements are irreversible and cause loss of information (Wittek, 2014).

Quantum gates are operators that act on qubits and are mathematically represented by unitary matrices. In the quantum realm, all operations, except measurements, are reversible. The relevant quantum gates for this work are the X gate (4), the Hadamard gate H (5), the Y rotation gate $R_y(\theta)$ (6), and the controlled-NOT gate $CNOT$ (7).

The X gate performs the following transformation: $|0\rangle \mapsto |1\rangle$ and $|1\rangle \mapsto |0\rangle$ (Wittek, 2014). Its matrix representation is defined as follows:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (4)$$

The Hadamard gate creates a uniform superposition of the basis states and is described by the matrix:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \quad (5)$$

When the H gate acts on a computational basis state $|k\rangle$, it transforms the input as follows: $H|k\rangle = \frac{1}{\sqrt{2}}(|0\rangle + (-1)^k|1\rangle)$.

The $R_y(\theta)$ gate modifies the probability of measuring a $|0\rangle$ or a $|1\rangle$ state by rotating a particular number of degrees around the y axis of the Bloch sphere; this is calculated using the matrix:

$$R_y(\theta) = \begin{pmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix}. \quad (6)$$

A handy gate for entangling (when paired with the H gate) is the controlled-NOT (CNOT) gate, a conditional gate that performs an X gate on the second qubit (target) if the state of the first qubit (control) is in state $|1\rangle$; otherwise the target qubit does not change. The control qubit remains unchanged regardless of the input values of any of the

qubits. The matrix representation of a CNOT gate is a 4×4 matrix defined as

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (7)$$

The CNOT gate is a generalized exclusive-or gate (XOR) gate that, in terms of the computational basis, performs the mapping: $|q_0, q_1\rangle \rightarrow |q_0, q_0 \oplus q_1\rangle$, where $\oplus$ is an XOR binary operator (Kasirajan, 2021).

Although quantum computing encompasses various operational models, this work focuses on the quantum circuit model. In a quantum circuit, qubits are represented by wires, and quantum gates are applied to these wires to manipulate the qubits. The qubits are ordered from top to bottom, with the most significant qubit at the top and the least significant at the bottom. Time progresses from left to right in the quantum circuit, with gates on the left applied before gates on the right. Operations that act on separate wires can be used in parallel. Quantum measurements are done to get the state of each wire or specific wires. Fig. 1 illustrates a 3-wire circuit with a depth 4. In this diagram, the leftmost layer corresponds to the first layer, and the top wire represents the most significant qubit, denoted as q_0 . The geometric shapes depict different types of gates, with each shape on a wire indicating a gate acting on the specified qubit q at that moment. The meter-like icons symbolize the measurement of one or more qubits, which concludes the quantum circuit. The circuit depth is determined by the minimum number of non-overlapping layers required, representing the sequence of gates in the circuit (Albornoz et al., 2021).

3. Quantum genetic algorithm for feature selection

The first method we propose consists of four stages: preprocessing, feature extraction, feature selection, and classification (see Fig. 2). During the preprocessing stage, the EEG signal is filtered to allow only the relevant frequencies for MI tasks, and wavelet packet decomposition (WPD) is used to analyze time-frequency features. A combination of hand-engineered features is extracted during the second stage. For the third stage, we propose a wrapper feature selection method based on a quantum genetic algorithm (QGA) designed for quantum processors. The classification stage is embedded in the wrapper method, where two classical machine learning algorithms are used. This approach draws inspiration from the work of Ramos and Vellasco (2018), but it differs from it in several aspects across each stage of the process. In the subsequent subsections, we will describe each step in detail.

3.1. Preprocessing and feature extraction

The initial stages of our BCI system involve signal preprocessing and feature extraction. The preprocessing stage aims to denoise and extract the essential components of the EEG signal. As the first step, we extract the time window corresponding to the imagery period for each trial.

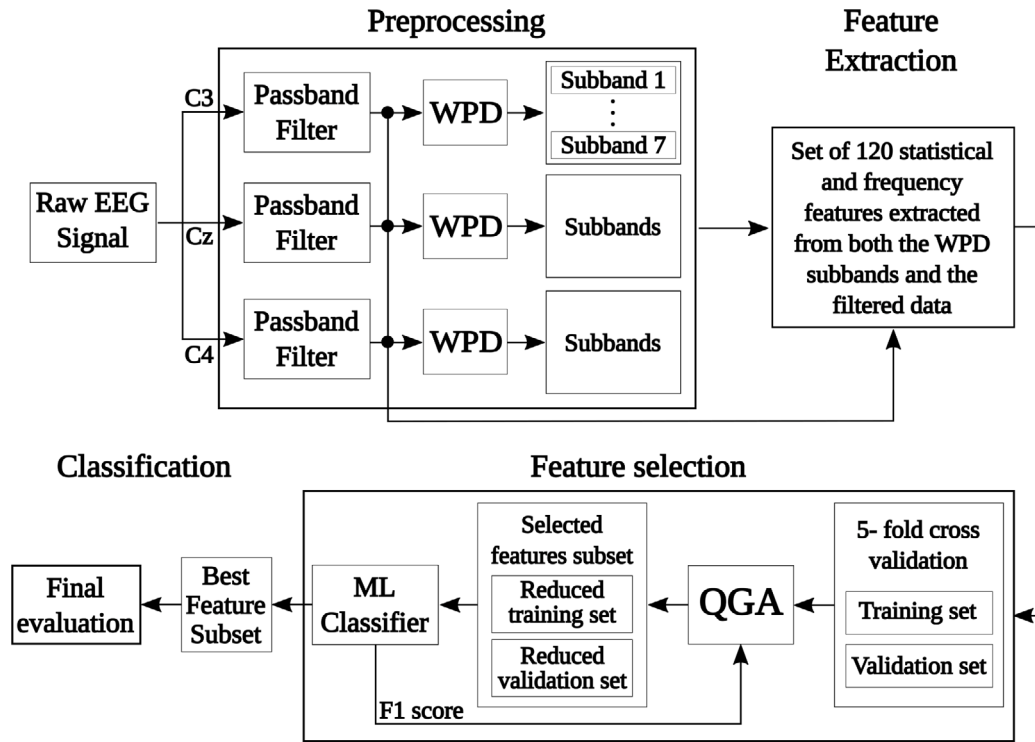


Fig. 2. ML approach based on a wrapper feature selection method based on the quantum genetic algorithm (QGA).

Given that in our evaluation dataset, this period lasted approximately 4 s with a sampling rate of 250 Hz, we extracted segments of 1000 time points.

MI-based BCIs use sensorimotor rhythms generated in the primary sensorimotor area of the brain cortex during motor imagery processes. The amplitude of these rhythms decreases during imagination or actual execution, which is called event-related desynchronization (ERD), and increases immediately after, which is named event-related synchronization (ERS) (Pfurtscheller & da Silva, 1999). In MI tasks, the main spectra of ERS and ERD are observed in the mu band (8–13 Hz) and beta band (13–30 Hz). The EEG-based BCI system interprets changes in the power spectrum of the mu and beta bands during MI as distinct control signals. Hence, as the second step, a sixth-order Butterworth bandpass filter from 4 to 32 Hz is applied to each channel to preserve the comprehensive information within these frequency bands.

The subsequent step involves conducting a time–frequency signal analysis through wavelet packet decomposition. The wavelet transform is highly effective in processing non-stationary and non-linear signals, such as EEG. WPD is a particular extension of the wavelet transform that offers enhanced flexibility in partitioning the time–frequency plane, yielding a highly detailed representation of local and global features at different frequency resolutions. Using a binary tree structure, the WPD method recursively divides the signal into subbands. The decomposition process is repeated until a desired level of decomposition is achieved, resulting in low-resolution and detailed components at each level derived from the preceding subband. These distinctive characteristics make WPD a powerful tool for discriminating MI-EEG signals (Taran & Bajaj, 2019; Zhang et al., 2017).

The wavelet basis selected for the decomposition is the commonly used fourth-order Daebuchies (db4) mother wavelet (Zhang, Zhang et al., 2015). The EEG signal is decomposed into five levels, resulting in thirty-two subbands covering a frequency range of 0 to 125 Hz, with a bandwidth of 3.9 Hz. To focus on the relevant frequency components in MI-EEG signals, only seven subbands are utilized, which are: 3.9–7.8 Hz, 7.8–11.7 Hz, 11.7–15.6 Hz, 15.6–19.5 Hz, 19.5–23.4 Hz, 23.4–27.3 Hz, 27.3–31.3 Hz.

The efficient extraction of discriminative features is critical to the performance of BCI systems for MI task classification. In the second stage of our BCI system, we extract a combination of statistical information features and subband energy features from the WPD coefficients. Additionally, we consider other features derived directly from the filtered signal. These feature extraction techniques have already been separately utilized in EEG research, and we integrate them to generate a diverse and comprehensive set of features.

The first five features that we selected are derived from WPD coefficients for each subband, and they are energy (Ting et al., 2008), volatility index (You et al., 2020), average nonlinear energy, variance of each subband (Vega-Escobar et al., 2015), and energy Ratio of adjacent subbands (Ramos & Vellasco, 2018). The next set of features was calculated from the filtered signal, and they are Hjorth parameters (Al-Qazzaz et al., 2021); skewness and kurtosis (Al-Qazzaz et al., 2021); first difference of time series (Chen et al., 2022); coefficient of variation (Chen et al., 2022); peak value of power spectral density (PSD) of theta, mu, and beta bands using the Welch method (Taran et al., 2018); and Burg's autoregressive spectral estimation method (Al-Fahoum & Al-Fraihat, 2014).

The computation of these parameters yields a feature vector comprising 120 features. The count is as follows: energy, volatility index, average nonlinear energy, and variance for each of the seven WPD subbands across the three channels create a total of 84 features; the energy ratio between adjacent WPD subbands over six sets of adjacent subbands for each of the three channels result in the generation of 18 additional features; mean values of Hjorth mobility, Hjorth complexity, skewness, kurtosis, first difference, and coefficient of variation between channels give a total of 6 features; peak value of PSD in the three frequency bands creates 3 additional features; finally, the parameters derived from the autoregressive model closes the count with 9 features.

3.2. Feature selection and classification

Due to the high dimensionality of the feature vector, feature selection is crucial to improve classification performance by excluding

redundant, irrelevant, or noisy features. The feature selection problem can be perceived as a combinatorial optimization problem homologous to the 0/1 knapsack problem, where the optimization metric is given by some properties obtained from applying a classification algorithm. Several QIEAs have been successfully applied to the knapsack problem, making them promising candidates for the feature selection problem.

We propose a wrapper method for feature selection based on the adaptive quantum genetic algorithm (AQGA), initially introduced by Wang et al. (2013), and subsequently implemented in a quantum circuit for circuit-based NISQ devices in a previous work (Rubio et al., 2021). The primary motivation behind this choice was to ensure compatibility with the intended execution on a Quantum Processing Unit (QPU). Notably, the existing literature features a scarcity of Quantum Evolutionary Algorithms (QEAs) explicitly tailored for QPUs since most proposals fall within the QIMs category. Hence, we decided to leverage one of the pre-existing algorithms designed with QPU execution in mind. As the article's purpose is an initial exploration of an algorithm primarily executable on a QPU rather than a concerted effort to surpass the capabilities of AQGA, our focus revolved around method proposal and component description, leaving ample room for further optimization.

Both works referenced the use of AQGA for solving numerical optimization problems, with its potential application to the feature selection problem remaining largely unexplored. During this stage, we undertake additional modifications to the AQGA, adapting (Rubio et al., 2021) implementation on Qiskit for IBM quantum devices to the PennyLane framework (Bergholm et al., 2020). In terms of computational complexity, while achieving a quadratic or any significant speedup is beyond the scope of this work, it is important to note that the complexity of our proposed method can be effectively compared to classical genetic algorithms, as we will illustrate in Section 3.2.5. This comparison is made feasible due to the inherent simplicity of the variational quantum circuits employed in our approach.

The algorithm is used to select the optimal feature vector that maximizes the F1 score of the classifier. The pseudo-code of the algorithm is presented in Algorithm 1. The details of the algorithm are explained in the following passages.

Algorithm 1 Adaptive quantum genetic algorithm

Input: Rotation and mutation tensors $\mathbf{R}', \mathbf{M}'$

Output: Best solution found p_{best}

```

1:  $t = 0$ 
2: Fill  $\mathbf{R}'$  with  $0.5\pi$ 
3: Execute  $\text{CIRCUIT}(\theta^t)$  for every  $r_{i,j}^t \in \mathbf{R}'$ 
4: Store the state  $s_k$  with the maximum sample count for every element in  $\mathbf{P}^t$ 
5: Evaluate  $\mathbf{P}^t$ 
6: Sort  $\mathbf{P}^t, \mathbf{R}^t, \mathbf{M}^t$  based on the fitness
7: Store the best individual of  $\mathbf{P}^t$  in  $p_{best}$ 
8: repeat
9:    $t = t + 1$ 
10:  Execute  $\text{CIRCUIT}(\theta^{t-1})$  for every  $r_{i,j}^{t-1} \in \mathbf{R}'$ 
11:  Store the state  $s_k$  with the maximum sample count for every element in  $\mathbf{P}^t$ 
12:  Evaluate  $\mathbf{P}^t$ 
13:  Sort  $\mathbf{P}^t, \mathbf{R}^t, \mathbf{M}^t$  based on the fitness
14:  Update the rotation angle,  $\theta_t \leftarrow \theta_{\max} - \frac{\theta_{\max} - \theta_{\min}}{g} \times t$ 
15:  Calculate the angle direction for every k-th qubit  $\omega_{i,j,k} \leftarrow \text{QRS}(\mathbf{P}_{i,j}^t, p_{best}, \mathbf{R}_{i,j}^{t-1}, \mathbf{M}_{i,j}^{t-1})$ 
16:  Update the rotation tensor  $\mathbf{R}^t \leftarrow \mathbf{R}^{t-1} + \omega * \theta_t$ 
17:  Update the mutation tensor  $\mathbf{M}^t$ 
18:  Apply disaster operation according to disaster condition
19:  Store the best individual of  $\mathbf{P}^t$  in  $p_{best}$ 
20: until  $t == g$ 

```

3.2.1. Representation

The AQGA uses a classical representation obtained by measuring a quantum register. The population $\mathbf{P}$ comprises I individuals, where each individual is a chromosome of N bits and represents a potential solution to the problem. For this application, N corresponds to the number of features ($N = 120$), and each bit represents one of those 120 features. The features with active values on the chromosome are selected to be part of the subset of features fed to the classifier.

In contrast to QIEAs that operate with probabilistic representations of qubit amplitudes, executing a quantum circuit causes the quantum information of the system to vanish, rendering it impossible to track the evolution of the individuals. Consequently, to implement the algorithm in the PennyLane framework, we rely on the quantum circuit representation introduced in Rubio et al. (2021).

In addition to the classical population, we maintain two key tensors to evolve the population while keeping the circuit depth as low as possible: the rotation tensor $\mathbf{R}$ and the mutation tensor $\mathbf{M}$. The rotation tensor stores the cumulative rotation angle for each qubit across generations. Using these angles as inputs, we execute quantum circuits at each generation to perform rotations. On the other hand, the mutation tensor consists of binary numbers that indicate whether a mutation operation is performed. A new cumulative rotation angle is generated when a mutation occurs, requiring two dimensions per qubit to accommodate the rotation angles before and after the mutation operator. These matrices collectively enable efficient evolution of the population while minimizing circuit depth.

PennyLane can extract different types of measurement results from quantum devices. We aim to optimize the population by rotating qubits towards the ideal solution, transitioning them from the state $|0\rangle$ to state $|1\rangle$, or vice versa. Thus, we sample the Pauli-Z operator's expectation value $\langle \psi | \sigma_z | \psi \rangle$. The resulting expectations range from 1 to -1, corresponding to $|\psi\rangle = |0\rangle$ and $|\psi\rangle = |1\rangle$, respectively. The result of the measurement is the number of times each measurement outcome occurred. We set the classical bit to the most observed state.

Given the hardware constraints of NISQ devices, processing each individual requires multiple quantum registers to accommodate the 120 qubits corresponding to the feature set. As the execution time scales with the system size, we opt to work with 12-qubit quantum systems to strike a balance between computational resources and practical feasibility. Therefore, the rotation tensor $\mathbf{R}$ dimension becomes $I \times J \times n \times 2$, where I is the population size; J is the number of quantum registers needed to process the full features set, in our case $J = 10$; n is the number of qubits of each quantum register, in this case, $n = 12$; and 2 are the cumulative angles needed for the evolution.

3.2.2. Initialization

A quantum circuit is executed every time a quantum measurement is required. All qubits are initialized in state $|0\rangle$ at the beginning of a computation. Since a rotation of 0.25π in the y axis replicates the effect of the Hadamard gate when applied to the state $|0\rangle$, we replace the Hadamard operation for a $R_y(0.5\pi)$ rotation at the beginning of the algorithm to reduce the circuit depth. To generate the initial population, we initialize all first angles of the rotation tensor to 0.5π (line 2), execute the quantum circuit for each quantum register of each individual (line 3), and store the most observed state of each qubit in the classical population (line 4).

The next step is to evaluate the population by classifying the data according to the features selected by each individual and obtaining the F1 score (line 5). Although accuracy is typically used as an optimization metric in the literature, the F1 score is the harmonic mean of precision and sensitivity and, thus, a more representative metric of the classifier's performance. The F1 score represents the fitness value of the individuals, and the one with the highest F1 score corresponds to the best individual.

Following the evaluation, the population, rotation, and mutation tensors are sorted in descending order, and the best solution is saved in p_{best} , see lines 6–7. Subsequently, the population is updated using the mechanism described next.

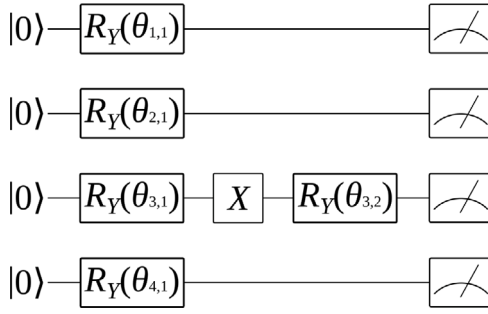


Fig. 3. Example of update of a quantum circuit of a 4-qubit individual.

3.2.3. Evolution

The following steps (lines 8–20) are iterated for a specific number of generations, denoted as g , to achieve the optimal solution. The initial action within this loop is to execute the quantum circuit for each quantum register in the population (line 10). The circuit consists of a maximum of three layers of quantum operations. The first layer applies R_Y gates to each qubit using the cumulative angles stored in R . This operation establishes the foundation of the circuit. The second layer represents the mutation operation, where X gates are applied to the qubits with an active value in the tensor M . This operation introduces variability in the population to improve the exploitation of the search space. Lastly, an additional set of R_Y gates is applied explicitly to the qubits that have undergone mutation, forming the third layer.

Fig. 3 demonstrates an example of the quantum circuit designed for a 4-qubit system to represent the quantum computation visually. While the maximum depth of the circuit is 3, it is worth noting that most qubits only require the application of a single operator due to the small mutation rate.¹ The gate complexity of this circuit scales linearly with the number of qubits, analogous to how the complexity of the crossover operator in a classical genetic algorithm (GA) depends on the length of the chromosome. For this reason, the execution of the quantum circuits used to evolve the population can be likened to the crossover and mutation operations in classical genetic algorithms in terms of complexity. As stated above, the output of the circuit is a dictionary with the count sample of the expectation values of the Pauli-Z operator for each qubit. Therefore, in the next step, we extract the state with the highest count value for each qubit and store it in the classical population P (line 11). We then evaluate and sort the population and auxiliary tensors in descending order according to their fitness value (lines 12, 13). We use the formula of the seminal work (Wang et al., 2013) to calculate the magnitude of the rotation angle according to the generation, see line 14.

One noteworthy distinction in our PennyLane implementation is the method employed to determine the position of the qubit in the Bloch sphere, which is crucial for determining the rotation direction. In contrast to the approach used in the Qiskit implementation (Rubio et al., 2021), which relies on the expectation values of the Pauli spin operators, we have adopted a different strategy to minimize the quantum load.

To avoid the need for additional quantum circuit executions solely for determining the position of the qubit, we leverage the cumulative angles stored in R to estimate its location in the Bloch sphere. Subsequently, we employ a quantum rotation scheme (QRS) to determine the direction of the angle (line 15) (Xiong et al., 2018). The QRS involves a lookup table where individuals from the population are compared to the best individual, and based on the position of the

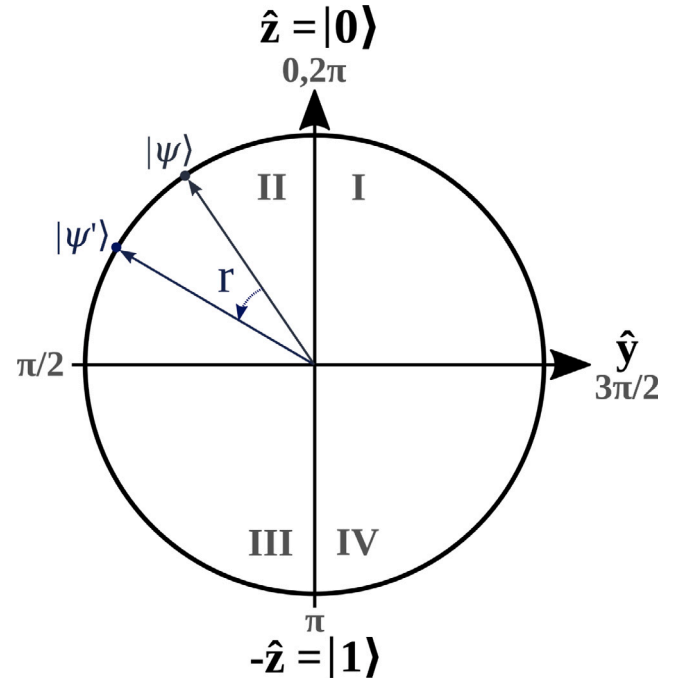


Fig. 4. Polar plot of the rotation strategy.

Table 1

Quantum rotation scheme.

$p_{i,j,k}$	$p_{best,j,k}$	$f(p_i) \geq f(p_{best})$	II/III	I/IV	$ 0\rangle$	$ 1\rangle$
0	1	0	+1	-1	± 1	0
0	1	1	-1	+1	0	± 1
1	0	0	-1	+1	0	± 1
1	0	1	+1	-1	± 1	0

qubits, the direction value ω is assigned as $\{0, +1, -1\}$. For our intended purposes, instead of tracking the position of the qubit by measuring the expectation value of the Pauli-Z operator, we use as a reference the current value of $\theta_{mod} = \theta_{i,j} \bmod 2\pi$. With this, we can track how close the qubit is to its poles and define four quadrants using as a reference the $|0\rangle$ state and counterclockwise: I, with $\theta_{mod} = [3\pi/2, 2\pi]$; II, with $\theta_{mod} = [0, \pi/2]$; III with $\theta_{mod} = [\pi/2, \pi]$; and IV with $\theta_{mod} = [\pi, 3\pi/2]$, see Fig. 4. The QRS we employ can be seen in Table 1.

After obtaining the magnitude and direction of the rotation angles, we update the tensor R by adding or subtracting these values to the cumulative angles (line 16). We then update the mutation tensor M by putting in 1 the qubits selected for mutation according to a mutation rate (line 17). Next, a disaster operation is conducted every time the population evolution reaches stagnation (line 18). This operation consists of replacing the last members of the population with new individuals initialized on an equal superposition state when the best individual stops improving for three generations. Finally, we update p_{best} with the best individual (line 19). We iterate these steps for g generations.

3.2.4. Classification

The classification is embedded in the wrapper selection method since the fitness is based on the classifier's performance. We used two popular traditional ML classifiers to compare the algorithm's performance: support vector machine (SVM) and random forest (RF). These two classifiers presented the best performance out of 179 classifiers evaluated in 121 datasets (Fernández-Delgado et al., 2014) and have been successfully applied to the classification of MI-EEG signals (Luo et al., 2016; Mebarkia & Reffad, 2019). The subset of features for each individual is fed to the classifier, and the F1 score is output to optimize the feature selection algorithm.

¹ Even further, due to the internal optimization of the PennyLane platform, the number of gates is reduced to one since our method do not include any conditional or entanglement operation.

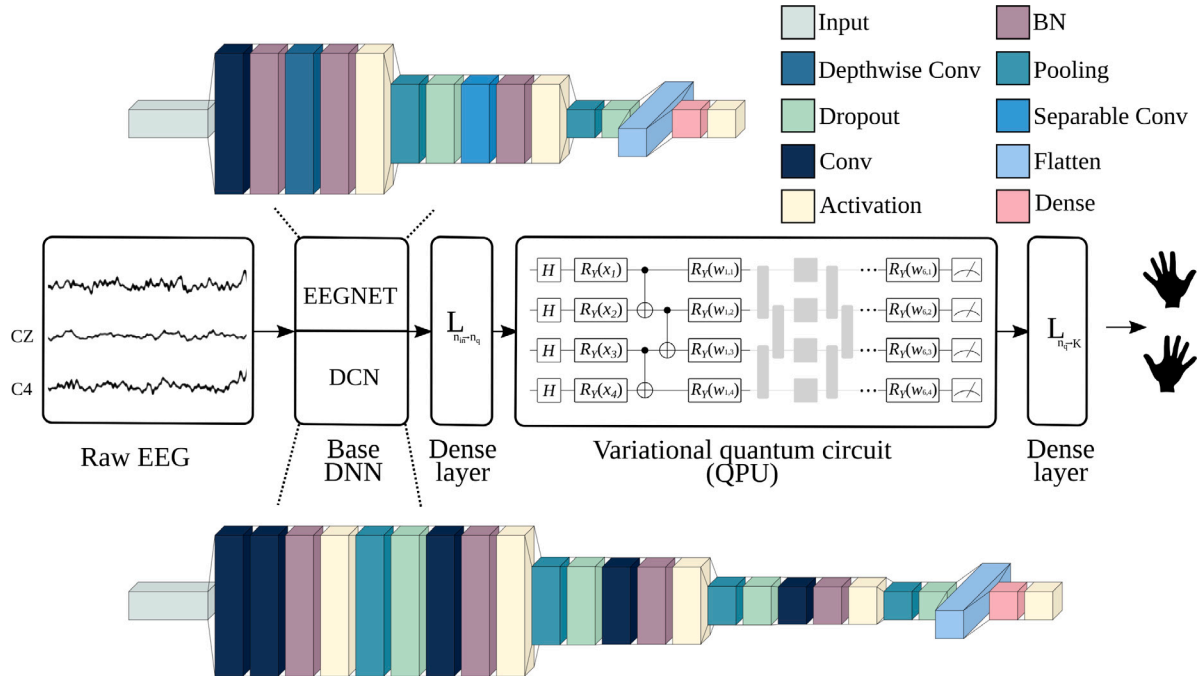


Fig. 5. Hybrid classical-quantum network for EEG-based MI classification. The base DNN and VQC can be interchanged for any appropriate network or quantum circuit.

3.2.5. Computational complexity analysis

The time complexity of our AQGA hinges on the configuration of certain parameters, including the quantum population size I , the maximum number of generations g , the required number of quantum registers J to process the complete feature set, the number of qubits n in each quantum register, and the complexity of the fitness evaluation function. Given that the complexity of the fitness function in this particular application depends on the classifier employed to evaluate the feature subsets, and our primary focus is solely on comparing the qualities of our quantum proposal against its classical counterpart, we will exclude this component from our analysis. Consequently, we can delve into the worst-case computational complexity of the proposed AQGA for feature selection by examining the key operations that contribute to the overall time complexity:

- Execute quantum circuit for each quantum register of each individual: $O(g \times I \times J \times n)$.
- Sort binary population, rotation, and mutation tensors: $O(g \times I \log I)$.
- Update rotation and mutation tensors: $O(g \times I \times J \times n)$.

The total complexity should be approximately $2 * O(g \times I \times J \times n) + O(g \times I \log I)$. Considering that the product of the number of generations and the size of each quantum register is greater than the size of the population, the first component should be dominant over the second one. Therefore, the overall execution time complexity of the proposed AQGA in the worst-case scenario is $O(g \times I \times J \times n)$. This can be contrasted with the time complexity of a generic classical GA, which is $O(g \times I \times N)$ (Samanta et al., 2017). In this particular implementation, we carefully addressed the constraints imposed by the limited number of qubits on NISQ devices, as discussed in Section 3.2.1. This implies that, for a GA, we have a chromosome length of N bits. However, for the AQGA, we partition these N qubits into J quantum registers, each containing n qubits, resulting in $N = J \times n$. This observation corroborates that, in the worst-case scenario, the complexity of AQGA is equivalent to that of GA.

4. Hybrid classical-quantum neural network

The second proposed method is an end-to-end hybrid classical-quantum network that combines a classical feature extraction network

with a quantum classification network. This approach automates the last three stages of our first proposed model, namely feature extraction, feature selection, and classification, without requiring specialized signal preprocessing. For the feature extraction part, we select five highly-cited and highly performing deep neural networks (DNNs) in EEG signal classification: Deep ConvNet (Schirrmeyer et al., 2017), EEGNet (Lawhern et al., 2018), EENet Fusion (Roots et al., 2020), TCNet-Fusion (Musallam et al., 2021), and ATCNet (Altaheri et al., 2023). For the classification part, we replaced the last fully connected layer of these DNNs with a variational quantum classifier (VQC) (Mari et al., 2020).

The term *hybrid* refers to the combination of classical and quantum computations in the model. In particular, the classical component includes the feature extractor, the preprocessing and postprocessing layers of the quantum classifier, and the weight optimizer of the PQC. The quantum part involves the variational quantum classifier executed on a quantum simulator for NISQ devices. Fig. 5 offers a graphical representation of the complete process of the hybrid model. The base architectures are used as feature extractors in the baseline DNN block. The details of the feature extraction and classification networks are described in the following subsections.

4.1. Preprocessing

Our proposed hybrid model operates on raw MI-EEG data, eliminating the need for complex EEG preprocessing. We extract information specifically from the time frame corresponding to the motor imagery period and standardize it per channel, ensuring it has a zero mean and unit variance. This standardization is based on the statistical properties of the training set.

To preserve the integrity of the data, we utilize the entire frequency band and include all EEG channels without employing any artifact removal techniques. Consequently, the input for the feature extractor is a matrix with dimensions $C \times T$, where C represents the number of electrodes, and T represents the number of time points. For the dataset used to evaluate our proposal, we have $C = 3$ channels and $T = 1000$ time points sampled at 250 Hz.

Table 2
Hybrid Deep ConvNet architecture.

Block	Layer	# filters	Size	Output	Activation	Options
1	Input			(3, 1000)		
	Reshape			(1, 3, 1000)		
	Conv2D	25	(1, 10)	(25, 3, 991)	Linear	Mode = valid
	Conv2D	25	(3, 1)	(25, 1, 991)	Linear	Mode = valid
	BatchNorm2d			(25, 1, 991)		Affine = false
	Activation			(25, 1, 991)	ELU	
	MaxPool2D		(1, 3)	(25, 1, 330)		Stride = (1, 3)
2	Dropout			(25, 1, 330)		$p = 0.5$
	Conv2D	50	(1, 10)	(50, 1, 321)	Linear	Mode = valid
	BatchNorm2d			(50, 1, 321)		Affine = false
	Activation			(50, 1, 321)	ELU	
	MaxPool2D		(1, 3)	(50, 1, 107)		Stride = (1, 3)
	Dropout			(50, 1, 107)		$p = 0.5$
	Conv2D	100	(1, 10)	(100, 1, 98)	Linear	Mode = valid
3	BatchNorm2d			(100, 1, 98)		Affine = false
	Activation			(100, 1, 98)	ELU	
	MaxPool2D		(1, 3)	(100, 1, 32)		Stride = (1, 3)
	Dropout			(100, 1, 32)		$p = 0.5$
	Conv2D	200	(1, 10)	(200, 1, 23)	Linear	Mode = valid
	BatchNorm2d			(200, 1, 23)		Affine = false
	Activation			(200, 1, 23)	ELU	
4	MaxPool2D		(1, 3)	(200, 1, 7)		Stride = (1, 3)
	Dropout			(200, 1, 7)		$p = 0.5$
	Flatten			(1400)		
Classification	Dense			4	Linear	
	VQC			4		Depth = 6
	Dense			2	Linear	

4.2. Feature extractor

Our hybrid system incorporates two baseline deep neural networks tailored explicitly for decoding EEG signals. This strategic choice allows us to explore the potential of these compact and generic CNNs, designed with little expert knowledge, in achieving competitive performance when combined with a quantum classifier for MI-EEG signal classification. Furthermore, in order to highlight the quantum classifier's capabilities, we have integrated three cutting-edge architectures into our hybrid network. These architectures are known for their exceptional performance in MI-EEG classification, and each is built upon the EEGNet framework. In the following sections, we will provide comprehensive details about these networks.

4.2.1. Deep ConvNet

In the study conducted by Schirrmeister et al. (2017), various deep CNN architectures were explored for decoding motor imagery tasks from raw EEG signals. Among these architectures, the 5-layer Deep ConvNet was utilized in this work. The Deep ConvNet comprises four blocks of convolutional and max-pooling layers, with the initial block specifically designed to handle the EEG input. This is followed by three standard convolution-max-pooling blocks and a fully connected softmax classification layer. In our hybrid method, the dressed quantum circuit replaces this last fully connected softmax classification layer.

A detailed description of the Hybrid Deep ConvNet model can be found in Table 2, where 3 represents the number of channels, and 1000 corresponds to the time points in the dataset used. The classification block will be discussed later. The first convolutional block comprises two distinct layers: a temporal convolution layer and a spatial convolution layer. While combining these layers into a single layer is theoretically possible since there is no activation function between them, employing two layers introduces regularization implicitly to the overall convolution process. This regularization arises from separating the linear transformation into a combination of temporal convolution and spatial filtering, contributing to the model's effectiveness.

The output of the convolutional layers undergoes batch normalization (Ioffe & Szegedy, 2015) before applying nonlinearity. Batch normalization standardizes the intermediate outputs by adjusting them to have zero mean and unit variance within each batch of training

data. It aims to improve training stability and generalization while accelerating convergence by maintaining layer inputs closer to a normal distribution during training.

Next, an ELU ($f(x) = \min(\exp(x) - 1, 0) + \max(x, 0)$) activation function (Clevert et al., 2016) is applied. The authors (Schirrmeister et al., 2017) conducted an analysis comparing the performance of the network using ReLU ($f(x) = \max(x, 0)$) activation functions instead and found that using ELU activation functions yielded superior results. Following the activation layer, a max-pooling layer is used to reduce the signal's sampling frequency. A dropout layer is added at the end of the block to prevent the coadaptation of different units. Dropout is applied to the inputs of all convolutional layers, excluding the firsts, with a probability of 0.5.

Blocks 2, 3, and 4 follow a similar structure to the first block, with the distinction that only a single convolutional layer is applied. The number of filters in each block is doubled, starting from 25 and ending with 200. The output of these layers is then flattened to serve as input for the dressed quantum circuit. The resulting feature vector comprises 1400 features.

4.2.2. EEGNet

The second network used as a feature extractor is the EEGNet, originally proposed by Lawhern et al. (2018). The EEGNet is a compact CNN architecture designed explicitly for EEG-based BCIs. It has demonstrated versatility across different BCI paradigms and has exhibited robust performance even when trained with limited data. The network consists of three primary operations. First, a 2D temporal convolution is used to learn frequency filters. Then, a depthwise convolution is employed to learn frequency-specific spatial filters. Lastly, a separable convolution is applied, which combines a depthwise convolution that learns a temporal summary for each feature map individually and a pointwise convolution that learns how to mix them optimally. Table 3 presents a detailed description of the Hybrid EEGNet.

Block 1 of the network comprises two convolutional layers designed to replicate the feature engineering approach of the filter-bank CSP algorithm (FBCSP) (Ang et al., 2012). A 2D convolutional layer is employed, where $F_1 = 8$ filters with dimensions (1, 125) perform convolutions over time. This generates eight feature maps that capture the EEG signal at various bandpass frequencies. By setting the temporal

Table 3
Hybrid EEGNet architecture.

Block	Layer	# filters	Size	Output	Activation	Options
1	Input			(3, 1000)		
	Reshape			(1, 3, 1000)		
	Conv2D	8	(1, 125)	(8, 3, 1000)	Linear	Mode = same
	BatchNorm2d			(8, 3, 1000)		
	DepthwiseConv2D	16	(3, 1)	(16, 1, 1000)	Linear	Mode = valid, depth = 2
	BatchNorm2d			(16, 1, 1000)		
	Activation			(16, 1, 1000)	ELU	
	MaxPool2D		(1, 5)	(16, 1, 200)		
	Dropout			(16, 1, 200)		$p = 0.5$
	Dropout			(16, 1, 200)		Mode = same
2	SeparableConv2D	16	(1, 16)	(16, 1, 200)	Linear	
	BatchNorm2d			(16, 1, 200)		
	Activation			(16, 1, 200)	ELU	
	MaxPool2D		(1, 10)	(16, 201, 20)		
	Dropout			(16, 1, 20)		$p = 0.5$
	Dropout			(16, 1, 20)		
	Flatten			320		
Classification	Dense			4	Linear	
	VQC			4		Depth = 6
	Dense			2	Linear	

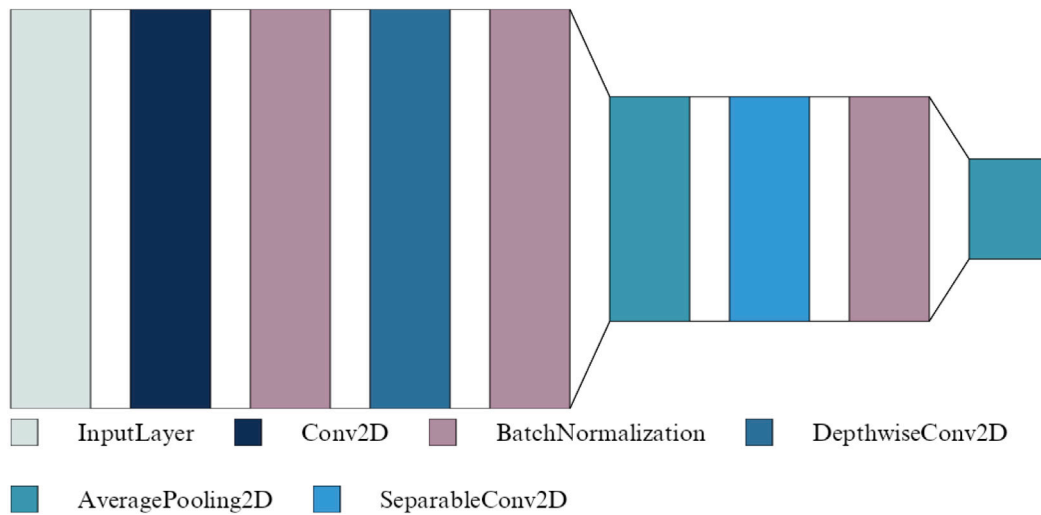


Fig. 6. Architecture of the EEGNet model up to the point just prior to the classification layer. For simplicity we omitted the activation, dropout, and flatten layers.

kernel length to half the sampling rate, frequency information at 2 Hz or higher can be effectively captured. Subsequently, a depthwise convolution (Chollet, 2017) convolves the data along the channel dimension to learn spatial filters. This operation provides a direct path to learn $D = 2$ spatial filters for each temporal filter, resulting in $F_1 \times D = 16$ filters, enabling efficient extraction of frequency-specific spatial filters. Both convolutions employ linear activation functions, omit the bias terms, and are followed by batch normalization. An ELU activation function is then applied. Dropout with a probability of 0.50 is utilized to regularize the model and prevent overfitting. Finally, a max-pooling layer of size (1, 5) is introduced to downsample the signal. While the original article employed average-pooling layers, superior results were achieved using max-pooling.

In block 2, a separable convolution is employed, consisting of a depthwise convolution of size (1, 16), followed by $F_2 = 16$ pointwise convolutions of size (1, 1) (Chollet, 2017). Separable convolutions offer several advantages, including parameter reduction and the ability to independently model relationships within and across feature maps. The separable convolution involves learning a kernel that summarizes each feature map individually and then optimally combining these feature maps with different weights. This operation mainly benefits EEG signals, as other feature maps can represent data at various time scales. Like Block 1, batch normalization, ELU activation, max-pooling of size (1, 10), and dropout layers are applied. Block 1 performs the feature extraction, while block 2 handles the feature selection. The

output of this block is flattened into a 320-feature vector, which serves as input for the subsequent quantum neural network (QNN) processing.

4.2.3. EEGNet fusion

The EEGNet Fusion, introduced by Roots et al. (2020), builds upon the EEGNet architecture. It incorporates three parallel branches, each processing the same input data in a distinct manner. These branches maintain the core structure of the EEGNet (see Fig. 6) while introducing variations in kernel sizes and the number of convolutional filters employed in the depthwise and separable convolutions for each branch. In particular, the convolutional layers of the initial block utilize kernel sizes of 4, 8, and 16, with filter dimensions of (1, 64), (1, 128), and (1, 256) for the three branches, respectively. For the separable convolutional layer in the subsequent block, a uniform kernel size of 16 is applied across all branches, accompanied by filter sizes of (1, 8), (1, 16), and (1, 32) for the three branches, respectively. Batch normalization, dropout, and activation layers follow the same methodology as in the EEGNet. Following the ultimate pooling layer, a feature fusion layer was implemented to combine the weights of the three CNN branches. The resultant output from this layer served as the input to the softmax classification layer to generate class probabilities.

4.2.4. TCNet-Fusion

The TCNet-Fusion model, proposed by Musallam et al. (2021), was based on the EEGNet and EEG-TCNet (Ingolfsson et al., 2020).

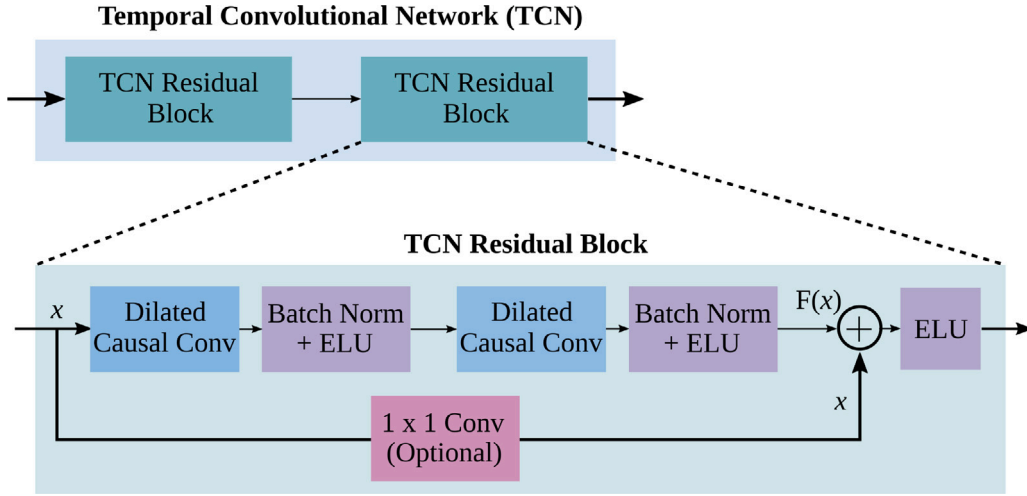


Fig. 7. Architecture of the TCN comprising two residual blocks.

The architecture consists of three key components. The first part is a convolutional block based on the EEGNet up to the output of the last pooling layer (see Fig. 6). The second part introduces a temporal convolutional network (TCN) for additional temporal features. The TCN used is the version proposed in Ingolfsson et al. (2020), featuring two residual blocks (see Fig. 7). These blocks include dilated casual convolutions, batch normalization, ELU activation functions, regular dropout, and 1×1 convolutions to maintain consistent depth shapes during the addition process. The third part is the concatenation of the TCN and the convolutional block outputs, mitigating feature loss and enriching feature maps. The final part flattens and concatenates the previously concatenated layer and the convolutional block outputs, forming a fused feature layer fed into a softmax activation function.

4.2.5. ATCNet

The ATCNet model (Altaheri et al., 2023) comprises three main components: convolutional block (CV), attention (AT) block, and temporal convolutional (TC) block. The CV block processes the MI-EEG signal similarly to the EEGNet architecture (Fig. 6). However, it deviates from EEGNet by employing a 2-D convolution in place of the separable convolution as well as distinct parameter values. The CV block produces a temporal sequence characterized by a more advanced representation, which is split into multiple windows and fed separately to the AT block and then to the TC block. The AT block highlights the most relevant information in the temporal sequence through the multi-head self-attention layer explained in Vaswani et al. (2017). Lastly, the TC block employs the same TCN as the TCNet-Fusion (see Fig. 7) to extract high-level temporal features within the sequence. The outputs from all windows are then combined and fed to a SoftMax classifier.

4.3. Variational quantum classifier

The concept behind hybrid classical-quantum training algorithms for variational circuits involves leveraging a quantum processor to evaluate an objective function $A(\mathbf{w})$ using a specific set of classical parameters $\mathbf{w}$. Subsequently, a classical algorithm is employed to optimize these parameters concerning the desired target by iteratively interacting with the quantum device (McClean et al., 2016). The entire routine is repeated until the target is minimized or maximized. This procedure is portrayed in Fig. 8. These algorithms aim to optimize the parameterized quantum circuits to generate specific wave functions or extract non-local information (Huang et al., 2021). This technique has the additional advantage of keeping the quantum circuit's depth low, therefore, mitigating noise (Cerezo et al., 2021).

Variational quantum circuits serve as quantum generalizations of feed-forward neural networks and are commonly referred to as variational quantum classifiers when applied to classification tasks. These circuits, also known as parameterized quantum circuits, are quantum algorithms that depend on a set of free parameters and follow the same structure as standard quantum circuits. Since our model manipulates classical data, the first step corresponds to a feature map that encodes the input data $\mathbf{x}$ into an n -qubit reference quantum state $|\mathbf{x}\rangle$, such as the vacuum or ground state. This is achieved by applying a unitary transformation to the initial state $|0\rangle^{\otimes n}$, denoted as a variational embedding layer:

$$\mathcal{E} : \mathbf{x} \rightarrow |\mathbf{x}\rangle = E(\mathbf{x})|0\rangle^{\otimes n}. \quad (8)$$

An example of these unitary gates is a single-qubit rotation parameterized by $\mathbf{x}$ (Jäger & Krems, 2023; Mari et al., 2020).

The next step involves transforming this initial state by applying an ansatz $U(\mathbf{w})$. In quantum algorithms, an ansatz generally refers to a subroutine of quantum operations that act on specific wires and are parameterized by $\mathbf{w}$ (Bai & Hu, 2023). When dealing with classification tasks, these parameters are optimized during training to minimize a predefined loss function $Loss(\mathbf{w})$. Subsequently, the trained model can be utilized to classify unseen data (Li & Deng, 2021). The layered gate architecture is among the three fundamental ansatz structures. A layer $\mathcal{L}$ can be defined as a unitary operation consisting of a sequence of gates that is repeated, acting on the input state $|\mathbf{x}\rangle$ to produce the output state $|\mathbf{y}\rangle$:

$$\mathcal{L} : |\mathbf{x}\rangle \rightarrow |\mathbf{y}\rangle = U(\mathbf{w})|\mathbf{x}\rangle. \quad (9)$$

An example of a widely used quantum layer contains a series of parameterized rotation gates applied to individual qubits, followed by a sequence of fixed entangling gates. A parameterized quantum circuit $\mathcal{Q}$ of depth D can be expressed as the product of D sequentially applied unitaries or as the concatenation of D variational layers $\mathcal{L}$:

$$\mathcal{Q} = \mathcal{L}_D \circ \dots \mathcal{L}_2 \circ \mathcal{L}_1 = \prod_{i=1}^D U_i(w_i) = U(\mathbf{w}), \quad (10)$$

with $U_i(w_i) = \exp(-iw_i V_i) W_i$. Here $\exp(-iw_i V_i)$ represents a rotation by an angle w_i induced by a Hermitian operator V_i satisfying $(V_i)^2 = I$. Additionally, W_i corresponds to an unparameterized quantum gate (Holmes et al., 2022). It is noteworthy that, unlike classical layers, quantum layers retain the dimensionality of the input states in the Hilbert space due to the inherent reversibility of quantum transformations (Huang et al., 2023).

Finally, a measurement of an observable $\hat{\mathbf{O}}$ is performed on the target quantum states to extract a classical output vector $\mathbf{y}$. The observable

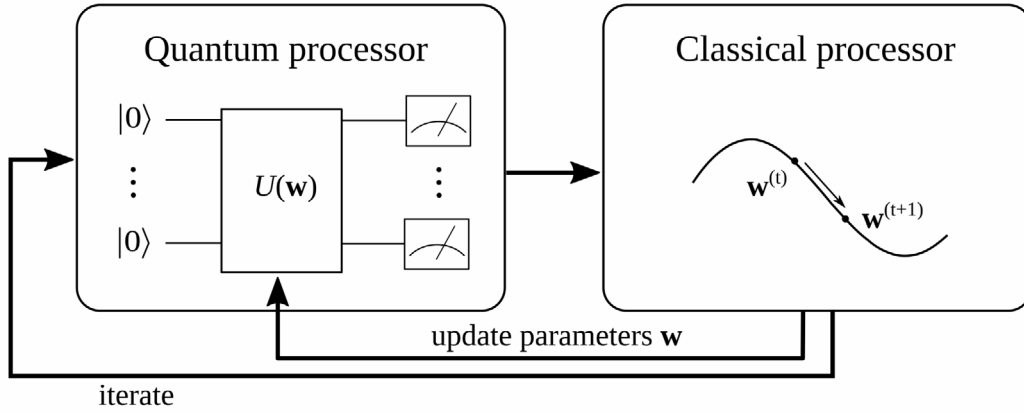


Fig. 8. Hybrid classical-quantum training process for variational circuits.

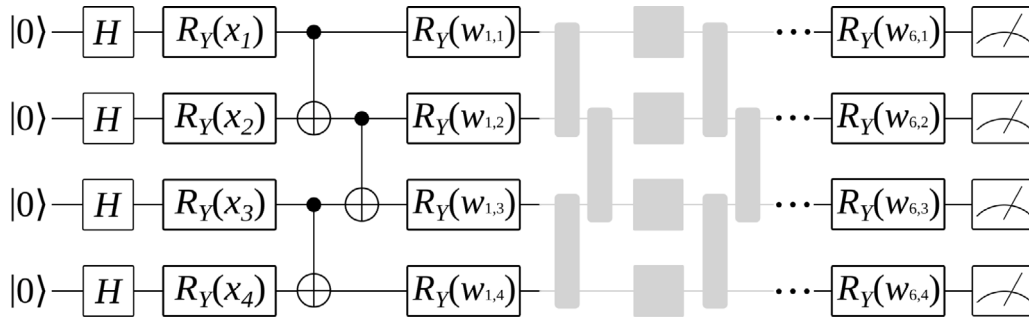


Fig. 9. VQC1: hand crafted variational quantum circuit based on Mari et al. (2020).

$\hat{\mathbf{O}}$ is composed of the expectation values of κ local observables, denoted as $\hat{\mathbf{O}} = [\hat{O}_1, \hat{O}_2, \dots, \hat{O}_\kappa]$. This measurement layer enables the mapping of a quantum state to a classical vector:

$$\mathcal{M} : |x\rangle \rightarrow \mathbf{y} = \langle x | \hat{\mathbf{O}} | x \rangle. \quad (11)$$

The complete quantum network, comprising both the initial embedding layer and the final measurement, can be expressed as

$$Q = \mathcal{M} \circ Q \circ \mathcal{E}. \quad (12)$$

We tested two VQCs for the quantum dressed network. The first circuit (VQC1) is based on Mari et al. (2020) and can be seen in Fig. 9; it has 4-qubit and a depth of 6 quantum layers. This specific configuration was selected through empirical analysis, as it demonstrated the most optimal performance among various alternatives. The specifications of this variational quantum classifier for each of the layers that conform the QNN are the following:

- Embedding layer $\mathcal{E}$: Amplitude embedding, where all qubits are initialized in superposition and then rotated according to the input parameters. This layer represents the set of Hadamard gates applied to each qubit, followed by the first set of R_y gates.
- Variational layers Q : A sequence of trainable rotation and constant entanglement layers are applied. A set of CNOT gates and a set of R_y gates represent one variational layer. The depth indicates how many of these layers our VQC will have.
- Measurement layer $\mathcal{M}$: For each qubit, the local expectation value of the Pauli-Z operator is measured.

The second VQC, dubbed VQC2, is based on the work by Acampora et al. (2023), where they used the VQC for the Wine dataset. We replaced the G gate with the one proposed in Schuld et al. (2020), described in Eq. (13), where $\alpha, \beta, \gamma \in [0, \pi]$.

$$G(\alpha, \beta, \gamma) = \begin{pmatrix} e^{i\beta} \cos \alpha & e^{i\gamma} \sin \alpha \\ -e^{-i\gamma} \sin \alpha & e^{-i\beta} \cos \alpha \end{pmatrix} \quad (13)$$

The specifications of this variational quantum classifier for each of the layers that conform the QNN are the following:

- Embedding layer $\mathcal{E}$: Amplitude embedding, provided by Penny-lane. This layer encodes a normalized 2^n -dimensional feature vector into the state of n qubits, in our case, $n = 4$.
- Variational layers Q : A sequence of trainable G and G -controlled gates are applied, as shown in Fig. 10.
- Measurement layer $\mathcal{M}$: We only measure the first qubit to get information about classes by measuring the probability of each computational basis state.

As previously mentioned, the number of features in each layer of the quantum network is constrained by the size of the physical system. In standard variational embedding layers, each classical element $\mathbf{x}$ is encoded into a single subsystem, resulting in the number of inputs being equal to the number of subsystems and outputs. However, to address this restriction, a dressed quantum circuit is proposed in Mari et al. (2020), outlined below.

It is necessary to perform preprocessing on the input data and postprocessing on the output data to establish a connection between the classical feature extractor and the VQC. This can be accomplished by incorporating a classical layer at the onset and culmination of the quantum network, resulting in the creation of a dressed quantum circuit $\tilde{Q}$:

$$\tilde{Q} = L_{n_{in} \rightarrow n_q} \circ Q \circ L_{n_q \rightarrow K}, \quad (14)$$

where Q is the associated quantum circuit defined in (12), and $L_{n_{in} \rightarrow n_q}$ and $L_{n_q \rightarrow K}$ are the preprocessing and postprocessing layers, given by the mathematical expression that describes a classical layer $\mathbf{x} \rightarrow \mathbf{y} = \phi(W\mathbf{x} + \mathbf{b})$. Here $\mathbf{x}$ and $\mathbf{y}$ are the input and output vectors, W and $\mathbf{b}$ are arbitrary real numbers that correspond to the weights and biases, and ϕ is a nonlinear function applied element-wise. The preprocessing layer maps the output of the feature extractor, with dimensions $n_{in} = 1400$

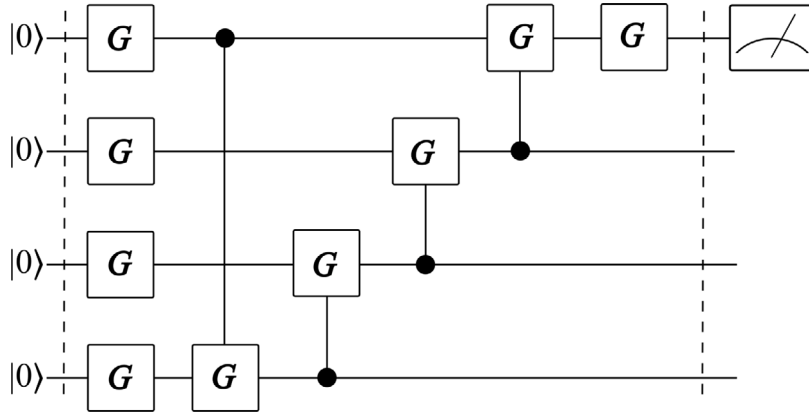


Fig. 10. VQC2: VQC based on Acampora et al. (2023) with modified gates from Schuld et al. (2020).

in the case of DeepConvNet or $n_{in} = 320$ in the case of EEGNet, to match the size of the VQC, which consists of $n_q = 4$ for the VQC1 and $n_q = 2^4 = 16$ for the VQC2. On the other hand, the postprocessing layer maps the 4-qubit output of the VQC1 (n_q) to a vector with $K = 2$ real outputs associated with the two classes of motor imagery tasks in our dataset: right-hand and left-hand. Since the output of the VQC2 is already in the desired format, the last postprocessing layer is omitted when using this classifier.

Simply put, the dressed quantum circuit implemented is a concatenation of:

- A classical linear preprocessing layer $L_{n_{in} \rightarrow n_q}$.
- A classical hyperbolic tangent activation function $\tanh$.
- A constant scaling of $\pi/2$.
- The variational quantum circuit Q shown in Fig. 9.
- A classical linear postprocessing layer $L_{n_q \rightarrow K}$.

This dressed quantum circuit is expressed in the classification block of Tables 2 and 3, where the linear layers represent the preprocessing and postprocessing layers, while the VQC denotes the variational quantum circuit VQC1 discussed in this section. Both the preprocessing layer and the VQC output a dimension of 4, which aligns with the number of qubits utilized in the circuit. On the other hand, the output of the postprocessing layer is 2, reflecting the number of classes involved in the classification task. In Fig. 5, the dressed quantum circuit is visually depicted within the last three blocks. The QPU block corresponds to the VQC, while the preceding and succeeding blocks represent the preprocessing and postprocessing layers, respectively.

5. Experimental results

In this section, we show several groups of experiments to evaluate the performance of the quantum approaches and assess their competitiveness against state-of-the-art techniques.

5.1. Dataset description

To evaluate the feasibility of the proposed methods, the dataset 2b from the BCI Competition IV, provided by Leeb et al. (2008), was used. This dataset encompasses two classes of MI-EEG recordings, corresponding to the imagery of right-hand movements (class 1) and left-hand movements (class 0). The EEG signals were captured from the C3, Cz, and C4 bipolar channels, as illustrated in Fig. 11, at a sampling frequency of 250 Hz. A bandpass filter ranging from 0.5 to 100 Hz was applied to the EEG signals, along with a 50 Hz notch filter to remove noise. The dataset also includes electrooculogram signals recorded with three monopolar electrodes, although only the EEG channels were utilized in these experiments.

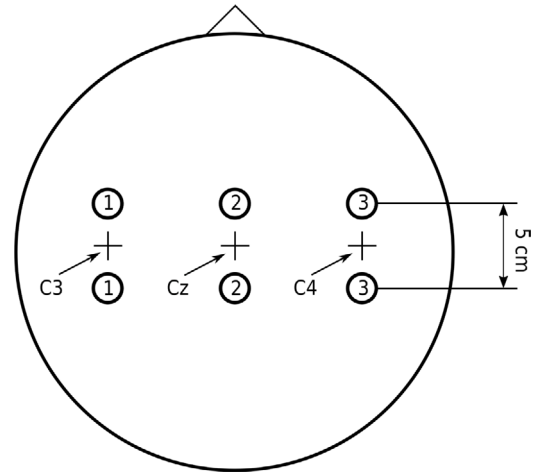


Fig. 11. Placement of the selected electrodes.

The dataset consists of 9 test subjects, with five sessions per subject, where the first two sessions contain training data without feedback, and the last three sessions were recorded with feedback. During the MI sessions without feedback, subjects performed 60 trials for each MI class, for a total of 120 trials. In each trial, the MI period is 3 s. In the MI session with feedback, subjects performed 80 IM task trials per class, with 160 trials in total. In each trial, the MI period is around 4 s. Each subject has a total of 720 tests, with some exceptions.

Since the relevant MI information occurs within a segment of approximately 4 s, subsets of signal data were extracted. Each trial consisted of 1000 data points, equivalent to a 4-s segment sampled at a frequency of 250 Hz. The EEG signals are represented as 2D matrices with the number of time points (1000) as the width and the number of electrodes (3) as the height.

5.2. Evaluation approaches

Our evaluation of the proposed models follows a subject-dependent approach, where we train and evaluate each model independently for each subject. Due to the high variability observed in MI-EEG signals across both sessions and subjects, various techniques have been developed for partitioning training and test sets in different datasets. In this study, we adopted two prominent methods documented in the literature.

The first method, called subject-dependent cross-validation, involves combining all sessions of a particular subject and then randomly

splitting it into a training set and a test set, maintaining an approximate 8:2 ratio. For parameter fine-tuning, we employed a 5-fold cross-validation approach within the training set. The second technique, named hold-out method, divides the training and test sets based on the different sessions, guided by the markers provided within the dataset. To enhance the reliability of our performance measurements and reduce variability, both of these methods are iterated five times. Subsequently, we computed the average results across these five iterations.

In addition, we also evaluated the end-to-end hybrid networks using a subject-independent approach. In the literature, a commonly employed technique for this approach is the cross-subject evaluation, known as leaving one subject out (LOSO). In LOSO, the model is trained and evaluated by several folds equal to the number of subjects; each fold uses a different subject for evaluation, while the remaining subjects are used for training. We adopt a similar concept for evaluating our networks. The data of a subject is appointed as the testing set and the data from the remaining subjects as the training set. However, we performed a 5-fold cross-validation on the training data for each subject to obtain robust performance estimates.

5.3. Experimental setup

Currently, the hardware limitations of NISQ devices hinder the performance of quantum algorithms. These limitations arise from factors such as cloud connectivity, the limited number of qubits, and inherent noise in NISQ configurations. Consequently, the fidelity of the quantum population's evolution and the quantum classifier's performance are affected. In light of these limitations, the AQGA and the QNN were executed on a quantum simulator of NISQ devices.

We used a server equipped with eight TITAN RTX 2080 GPUs with 24 GB of memory for these experiments. The algorithms were implemented in Python, using the PennyLane (Bergholm et al., 2020) framework for programming the quantum mechanisms. PennyLane is a cross-platform Python library for the differentiable programming of quantum computers. When using the PennyLane PyTorch and PennyLane TensorFlow plugins, the quantum circuit can seamlessly integrate as a layer within PyTorch or Keras. This quantum layer is then combined with classical layers. Consequently, the built-in loss functions and optimizers provided by PyTorch and Keras can be leveraged to update the parameters efficiently. The model is trained and evaluated for each subject using a single GPU. However, to leverage the computational power of the servers, the experiments are run in parallel for different subjects on different GPUs.

For all experiments, we used the following training configurations. The parameters of the feature selection algorithm were set to 40 generations, a population size of 50, a mutation rate of 2%, a disaster rate of 20% after the population stops improving for three generations, for the self-adjusting angle $\theta_{max} = 0.35\pi$ and $\theta_{min} = 0.01\pi$. As mentioned, we use ten quantum registers of 12 qubits to evolve the whole individual. The results presented are the average of 10 runs. For the cross-validation approach, the end-to-end models were trained using the Adam optimizer with an initial learning rate of 0.001, a decay rate of 0.9 with a step size of 15, a batch size of 128, and a cross-entropy loss over 300 epochs and 200 epochs for the subject-dependent and the subject-independent classification respectively. The QNN used in this set of experiments is the VQC1, operating with four qubits, six quantum layers, and an initial spread of random quantum weights of 0.01.

For the hold-out approach, the hybrid models were trained using the Adam optimizer with a learning rate of 0.0009, a batch size of 64, and a categorical cross-entropy loss over 300 epochs with a patience of 100. For this set of experiments, the classification was performed using the VQC2, operating with four qubits and an initial spread of random quantum weights set to 0.01. These hyperparameters were determined empirically through several experiments to aid the models to generalize well.

5.4. Performance metrics

The proposed models in this work are evaluated using the accuracy (Eq. (15)), the F1 score (Eq. (16)), and Kappa score (Eq. (17)). In binary classification, these metrics rely on the following values: true positives (TP) for correctly predicted positive elements, true negatives (TN) for correctly predicted negative instances, false positives (FP) for negative instances incorrectly classified as positive, and false negatives (FN) for positive instances predicted as negative. Accuracy denotes the proportion of correctly classified instances within the test set and is calculated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (15)$$

While accuracy alone may not provide a complete picture of the model's performance, additional metrics can provide valuable insights. Precision measures the fraction of correctly classified positive instances out of all positive predictions, while recall measures the proportion of correctly classified positives out of all actual positives. The F1 score combines precision and recall into a single metric, offering a more accurate measure of the model's performance. It is calculated as follows:

$$F1 = \frac{2TP}{2TP + FP + FN} \quad (16)$$

As previously mentioned, the F1 score was utilized to optimize the QGA in these experiments.

The Kappa value is an evaluation metric used to assess classification performance by measuring the level of agreement between the model's predictions and actual labels while accounting for random effects. A higher Kappa value indicates superior classification performance, while a lower score reveals the model's vulnerability to bias in the presence of imbalanced data. It is defined as follows:

$$\text{Kappa} = \frac{P_0 - P_e}{1 - P_e} \quad (17)$$

where P_0 is the actual probability of agreement, and P_e is the expected probability of chance agreement.

During our subject-dependent training, we employed p-values to assess the statistical significance between our proposed hybrid network and other baseline models that shared a similar training methodology. Additionally, in the subject-independent training phase, we utilized p-values to analyze the statistical differences between the best performing state-of-the-art model and the classical and hybrid networks we implemented. To achieve this, we employed the paired T-test, a widely used statistical tool.

The p -value plays a crucial role in hypothesis testing. It quantifies the strength of evidence against the null hypothesis, which assumes that there is no significant difference between the compared models. A p -value greater than 0.05 indicates no statistically significant difference in performance between the two models; a p -value less than 0.05 (denoted by *) suggests a statistically significant difference, meaning that the observed disparities are unlikely to have occurred by chance alone, providing evidence to reject the null hypothesis; and a p -value less than 0.01 (denoted by **), signifies an exceptionally high level of statistical significance.

By incorporating the paired T-test and interpreting the associated p-values, we effectively gauged the statistical significance of the performance differences between our proposed hybrid network and the other baseline models. This rigorous analysis allows us to draw robust and reliable conclusions, enhancing the credibility and impact of our research outcomes.

5.5. Performance comparison: subject-dependent classification

5.5.1. Cross-validation approach

Table 4 summarizes the results, showcasing the average test F1 scores across all iterations and their corresponding standard deviations

Table 4

Performance (mean F1 score and standard deviation) comparison of subject-dependent cross-validation classification for the proposed methods (VQC1) with baseline models.

	QGA+SVM	QGA+RF	EEGNet	Hybrid EEGNet	Deep ConvNet	Hybrid Deep ConvNet
B1	0.7999 $\pm$ 0.0015	0.8049 $\pm$ 0.0022	0.8741 $\pm$ 0.0115	0.8611 $\pm$ 0.0210	0.8612 $\pm$ 0.0092	0.8089 $\pm$ 0.0283
B2	0.6439 $\pm$ 0.0015	0.6237 $\pm$ 0.0058	0.6508 $\pm$ 0.0201	0.6753 $\pm$ 0.0183	0.6112 $\pm$ 0.0259	0.6268 $\pm$ 0.0102
B3	0.6050 $\pm$ 0.0158	0.5907 $\pm$ 0.0028	0.7633 $\pm$ 0.0038	0.7680 $\pm$ 0.0082	0.7528 $\pm$ 0.0152	0.7783 $\pm$ 0.0309
B4	0.9106 $\pm$ 0.0026	0.9341 $\pm$ 0.0012	0.9757 $\pm$ 0.0053	0.9830 $\pm$ 0.0071	0.9702 $\pm$ 0.0052	0.9624 $\pm$ 0.0057
B5	0.8303 $\pm$ 0.0025	0.8625 $\pm$ 0.0019	0.9302 $\pm$ 0.0118	0.9228 $\pm$ 0.0232	0.9173 $\pm$ 0.0101	0.9069 $\pm$ 0.0300
B6	0.7987 $\pm$ 0.0039	0.8069 $\pm$ 0.0039	0.7761 $\pm$ 0.0112	0.7790 $\pm$ 0.0116	0.7871 $\pm$ 0.0113	0.7978 $\pm$ 0.0322
B7	0.7429 $\pm$ 0.0041	0.7488 $\pm$ 0.0045	0.8398 $\pm$ 0.0137	0.8456 $\pm$ 0.0170	0.7254 $\pm$ 0.0165	0.7598 $\pm$ 0.0667
B8	0.7867 $\pm$ 0.0030	0.7997 $\pm$ 0.0022	0.8555 $\pm$ 0.0108	0.8598 $\pm$ 0.0075	0.7806 $\pm$ 0.0093	0.7862 $\pm$ 0.0359
B9	0.8047 $\pm$ 0.0065	0.8097 $\pm$ 0.0015	0.8603 $\pm$ 0.0139	0.8549 $\pm$ 0.0141	0.8390 $\pm$ 0.0088	0.8282 $\pm$ 0.0176
Average	0.7692	0.7757	0.8362	0.8388	0.8050	0.8062
SD	0.0885	0.1023	0.0907	0.0846	0.1010	0.0886

The bold font highlights the best results among different models.

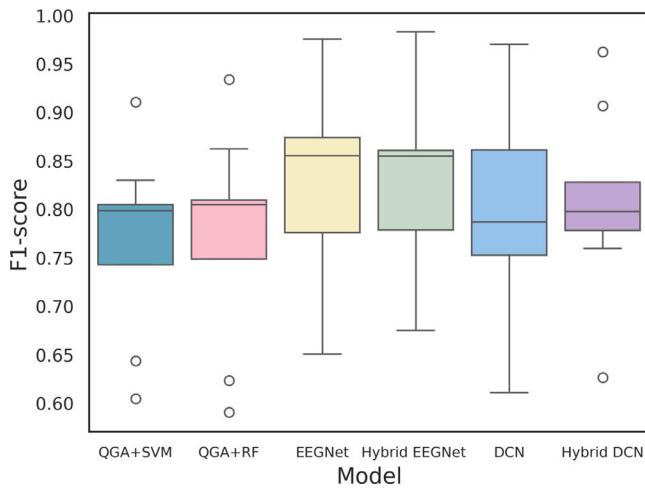


Fig. 12. Box plot of the F1-scores for each model across all subjects.

achieved through the implemented methods during subject-dependent cross-validation training. Both classical and hybrid deep learning models outperform the proposed machine learning method in terms of F1 scores. Notably, the machine learning methods exhibit a consistent and low intra-subject standard deviation despite their lower F1 scores, signifying enhanced stability between runs for each specific subject.

In contrast, the hybrid EEGNet model delivered the highest mean F1 score (0.8388) and demonstrated the least variability across all subjects, as indicated by its low inter-subject standard deviation (0.0846), depicted in Fig. 12. This box plot provides a graphical representation of data distribution and facilitates model comparisons. Remarkably, the Hybrid EEGNet showcases the most desirable distribution within the proposed models with no outliers. It is important to consider that due to the computational demands associated with hybrid training, driven by hardware limitations of NISQ devices, hyperparameter tuning can be particularly challenging. Nevertheless, the performance of the hybrid models closely resembles that of classical networks, indicating that the variational quantum circuit VQC1 behaves in a manner similar to a fully connected softmax layer.

Table 5 compares the accuracy performance between the hybrid EEGNet network and various state-of-the-art methods. To ensure a fair evaluation, we have specifically chosen a selection of the most recent and highly regarded end-to-end architectures proposed in the field of MI that adopted a similar methodology for partitioning the training and test sets. These models are the hybrid-scale CNN (HS-CNN) (Dai et al., 2020), the EEG-inception (Zhang et al., 2021), the parallel CNN (PCNN) (Han et al., 2022), the multi-scale time–frequency block CNN (MTFB-CNN) (Li, Chen et al., 2023), the compact multi-branch 1D

CNN (CMO-CNN) (Liu et al., 2023), and the multi-scale hybrid CNN (MSHCNN) (Tang et al., 2023). The primary criterion for comparison in the table is accuracy, which serves as a robust measure of performance. The results in the table are the same reported in their seminal works.

While a significant portion of NISQ algorithm research has traditionally focused on harnessing quantum hardware for toy examples or computer vision problems, it has often overlooked real-world applications. We aim to enhance the scope of quantum hardware's potential by employing hybrid algorithms in a broader range of problems, such as EEG signal classification, to improve MI-BCI performance. Even though our investigation did not reveal any quantum advantage or improvement, our findings underscore the competitive performance of our hybrid EEGNet model. Although it exhibited a statistically significant difference (p -value=0.006) compared to the top-performing model (MTFB-CNN), there was no statistically significant difference between the hybrid EEGNet and half of the baseline models used for comparison (HS-CNN, PCNN, MSHCNN).

It is essential to emphasize that our proposed hybrid models were not expressly designed to address the limitations of advanced deep learning models for EEG-based MI classification or to mitigate challenges associated with VQCs, including trainability and generalization. Our models are constructed using compact and generic CNNs and a general-purpose ansatz (VQC1). Many recent proposals in the literature that achieve high classification accuracy in MI-EEG classification are based on the EEGNet architecture, incorporating additional blocks and strategies for performance enhancement (e.g., ATCNet, TCNet-Fusion, EEGNet-Fusion). Consequently, it is unrealistic to expect baseline CNNs like EEGNet and DeepConvNet, and hybrid models based on them, to surpass cutting-edge architectures.

However, despite these constraints, our hybrid model delivered results on par with state-of-the-art approaches in terms of accuracy. Fig. 13 presents a line graph depicting the accuracy results (in %) for each subject, as achieved by the different models under comparison. This visual representation highlights variations in accuracy across individual subjects within each model, with our proposed model performing very similar than most of the state-of-art proposals. While the PCNN exhibited the lowest mean accuracy, it also demonstrated the lowest standard deviation. As evident in the line graph, its performance stands out with a notably distinct behavior compared to the other models.

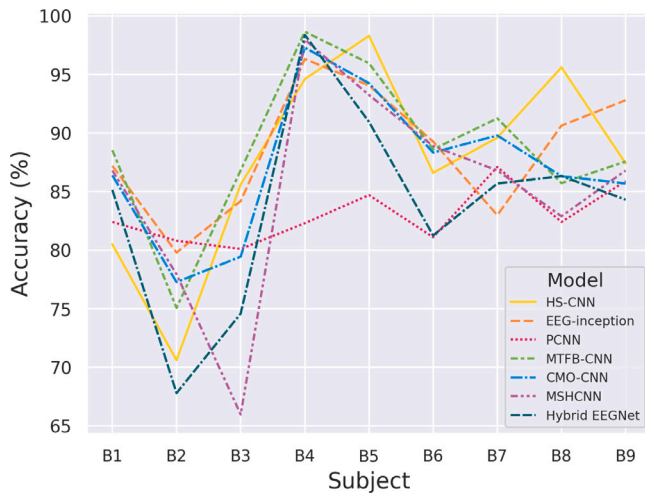
Ultimately, utilizing domain-specific deep learning models for MI-EEG signal classification, combined with a tailored ansatz designed for this specific application, holds promise for achieving superior results, even when facing constraints imposed by NISQ devices. Additionally, a systematic approach to hyperparameter fine-tuning and the incorporation of preprocessing techniques like data augmentation can substantially enhance the network's performance.

Table 5

Performance (% accuracy) comparison of subject-dependent cross-validation classification for the proposed Hybrid EEGNet with cutting-edge models.

	HS-CNN	EEG-inception	PCNN	MTFB-CNN	CMO-CNN	MSHCNN	Hybrid EEGNet
B1	80.5	87.2	82.4	88.54	86.38	86.8	85.14
B2	70.6	79.79	80.8	75.07	77.27	77.94	67.79
B3	85.6	84.19	80.1	86.81	79.44	65.97	74.58
B4	94.6	96.32	82.3	98.65	97.29	97.97	98.38
B5	98.3	94.06	84.7	95.95	94.25	93.24	90.95
B6	86.6	89.27	81.1	88.61	88.33	88.88	81.25
B7	89.6	82.98	87.1	91.25	89.79	86.80	85.69
B8	95.6	90.63	82.4	85.72	86.31	82.89	86.32
B9	87.4	92.8	85.9	87.57	85.69	86.80	84.31
Average	87.64	88.58	83.0	88.69	87.19	85.25	83.82
SD	8.0	5.19	2.27	6.30	5.98	8.67	8.32
p-value	0.064	0.024*	0.761	0.006**	0.018*	0.460	

The bold font highlights the best results among different models.

**Fig. 13.** Accuracy distributions of the compared methods.

5.5.2. Hold-out approach

For the second set of experiments, we conducted training on the hybrid models, employing the original competition division of the training and the testing sets, and integrating the VQC2 within the dressed quantum circuit. In addition to the baseline CNNs used in the first set of experiments (EEGNet and Deep ConvNet), we incorporated three additional encoder networks known for their state-of-the-art performance in multiclass MI-EEG classification: ATCNet, TCNet-Fusion, and EEGNet-Fusion.

Table 6 presents the mean test F1 scores across all iterations and their corresponding intra-subject and inter-subject standard deviations, showcasing the results achieved through these methods during hold-out subject-dependent training. It is worth noting that in its seminal work, the ATCNet model outperformed both EEGNet and TCNet-Fusion. Therefore, it is reasonable to expect that when used as a feature extractor and combined with a quantum classifier, it will also outperform the other architectures employed as encoders. The hybrid ATCNet obtained the highest mean F1-score (0.8551) and the lowest inter-subject standard deviation (0.0991) among all tested architectures.

We observe a slight decrease in the overall performance of the hybrid EEGNet and hybrid Deep ConvNet models in this set of experiments compared to the cross-validation approach. This decrease can be attributed to the specific nature of the approach, where the datasets for individual subjects are relatively small and exhibit some variability between sessions with and without feedback. It is noteworthy that these two models performed better when using the VQC1.

In Fig. 14, we visually express the average results in terms of F1-score, Kappa value, and accuracy (expressed as a decimal percentage)

across all subjects achieved by the hybrid models. The black line positioned above each bar represents the standard deviation among different subjects. We can appreciate that the best performance in terms of all three metrics is demonstrated by the Hybrid ATCNet, closely followed by the Hybrid TCNet-Fusion. As discussed in Section 5.5.1, using a deep learning architecture designed to enhance EEG-based MI classification yields superior results. This underscores the idea that simply replacing the classifier of a baseline CNN with a quantum classifier may not necessarily result in superior performance.

Table 7 summarizes the accuracy performance between the hybrid ATCNet model and the overall accuracy of various state-of-the-art methods employing the hold-out approach. These models are the Double-branch Graph Convolutional Attention Neural Network (DGCAN) (Ma et al., 2023), the multiscale Siamese convolutional neural network with cross-channel fusion (MSCCF-Net) (Shen et al., 2022), WT-Isomap (Li et al., 2017), Boltzmann Machine (Lu et al., 2017), TCNet-Fusion (Musallam et al., 2021), and ATCNet (Altafari et al., 2023). The primary focus of this comparison is accuracy, a robust measure of performance. The results presented in the table remain consistent with those reported in their seminal works, except for TCNet-Fusion and ATCNet, which represent the outcomes obtained from running the original models using the specific dataset and settings employed in this study.

Based on these results, there is no statistically significant difference in accuracy between the hybrid ATCNet and the other models. Moreover, both the classical ATCNet and the hybrid ATCNet exhibit some of the lowest standard deviations, measuring 9.21 and 9.78, respectively. This trend is surpassed only by the MSCCF-Net model, which achieves the highest mean accuracy at 87.33% and the lowest standard deviation at 7.97. Fig. 15 illustrates the kappa distributions across subjects for the two top-performing hybrid models, Hybrid ATCNet and Hybrid TCNet-Fusion, alongside the most recent proposal documented in the literature, DGCAN. Notably, both hybrid models exhibit a consistent performance across all subjects and showcase more uniform distributions compared to the classical model.

It is crucial to emphasize that by enhancing the encoder network, the overall performance of the hybrid network is likewise improved. This finding reaffirms the potential of the variational quantum classifier as a promising method in the realm of BCIs. Furthermore, it serves as evidence of the broader applicability of NISQ algorithms, extending their reach beyond the confines of laboratory settings.

5.6. Performance comparison: subject-independent classification

Table 8 provides a comparative analysis of the accuracy performance across various models used in the study. It includes the hybrid models utilized in the first set of experiments, Hybrid EEGNet and Hybrid Deep ConvNet, as well as the top-performing hybrid models employed in the second set of experiments, Hybrid ATCNet and Hybrid

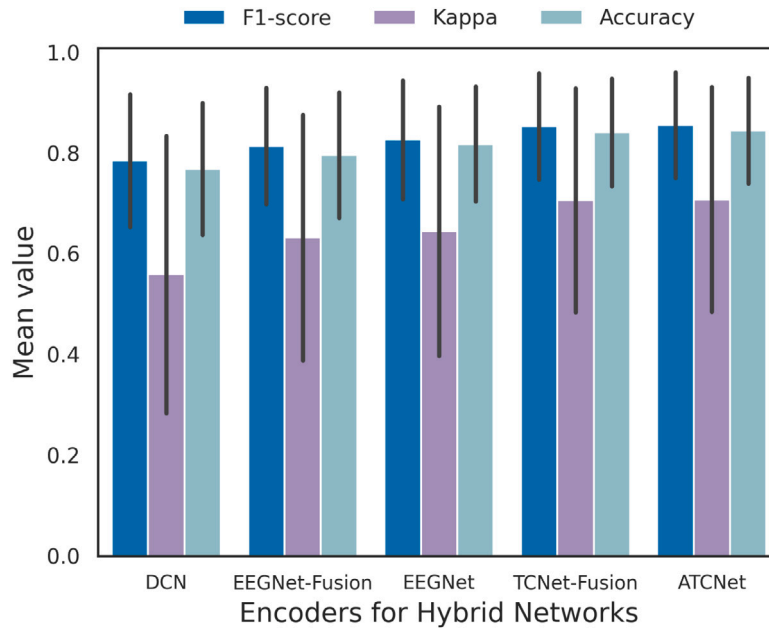


Fig. 14. Histograms of the classification results (mean F1-score, kappa value, and accuracy) based on the hybrid models.

Table 6

Performance (mean F1 score and standard deviation) comparison of hold-out approach for the proposed dressed quantum network (VQC2) using different feature extraction networks.

	Hybrid EEGNet	Hybrid Deep ConvNet	Hybrid ATCNet	Hybrid TCNet-Fusion	Hybrid EEGNet Fusion
B1	0.6602 ± 0.0540	0.6459 ± 0.0904	0.7618 ± 0.0097	0.7251 ± 0.0170	0.6817 ± 0.0055
B2	0.6235 ± 0.0068	0.5479 ± 0.1065	0.6431 ± 0.0070	0.6568 ± 0.0231	0.6049 ± 0.0089
B3	0.8466 ± 0.0077	0.7944 ± 0.0825	0.8323 ± 0.0078	0.8332 ± 0.0039	0.8039 ± 0.0347
B4	0.9823 ± 0.0059	0.9614 ± 0.0053	0.9865 ± 0.0039	0.9833 ± 0.0015	0.9844 ± 0.0026
B5	0.7936 ± 0.0156	0.7104 ± 0.0113	0.9479 ± 0.0140	0.9321 ± 0.0142	0.8035 ± 0.0453
B6	0.8446 ± 0.0227	0.7959 ± 0.0156	0.8250 ± 0.0104	0.8360 ± 0.0068	0.8297 ± 0.0062
B7	0.8812 ± 0.0377	0.8719 ± 0.0203	0.9031 ± 0.0111	0.9208 ± 0.0059	0.9052 ± 0.0014
B8	0.9365 ± 0.0015	0.9208 ± 0.0064	0.9250 ± 0.0051	0.9271 ± 0.0059	0.9018 ± 0.0053
B9	0.8649 ± 0.0403	0.8127 ± 0.0093	0.8708 ± 0.0078	0.8603 ± 0.0121	0.8081 ± 0.0083
Average	0.8259	0.7846	0.8551	0.8527	0.8137
SD	0.1131	0.1244	0.0991	0.0994	0.1089

The bold font highlights the best results among different models.

Table 7

Performance (% accuracy) comparison of hold-out approach for the proposed Hybrid ATCNet with top performing methods.

	WT-Isomap	Boltzmann Machine	TCNet Fusion ^a	ATCNet ^a	MSCCF-Net	DGCAN	Hybrid ATCNet
B1	84.6	81.0	75.31	78.85	77.81	84.2	76.46
B2	66.3	65.0	66.67	64.88	70.35	77.84	64.40
B3	62.9	66.0	85.21	84.48	86.56	92.89	83.23
B4	95.8	98.0	97.92	98.54	97.81	89.87	98.65
B5	89.2	93.0	96.35	89.27	92.50	98.36	94.79
B6	97.9	88.0	87.50	83.65	89.38	72.51	82.60
B7	82.1	82.0	91.15	91.98	91.56	95.85	90.31
B8	86.3	94.0	98.02	92.5	93.13	95.86	91.5
B9	97.1	91.0	87.71	88.54	86.88	63.92	87.08
Average	84.69	84.22	87.32	85.85	87.33	85.71	85.56
SD	11.99	11.25	9.99	9.21	7.97	11.29	9.78
P-value	0.840	0.629	0.054	0.614	0.097	0.95	

The bold font highlights the best results among different models.

^a Reproduced.

TCNet-Fusion. These models are benchmarked against one of the most recent proposals in the literature for subject-independent classification, MSHCNN (Tang et al., 2023). Additionally, we compare the classical Deep ConvNet and EEGNet.

The hybrid EEGNet exhibited the best performance out of all the hybrid models tested, achieving an accuracy of 73.72%, followed by

the hybrid Deep ConvNet model, which obtained an average accuracy of 72.48%. Furthermore, both the hybrid Deep ConvNet and the hybrid EEGNet models achieved a higher average accuracy than their classical counterparts. They showed no statistically significant difference when compared to the MSHCNN model, with p-values of 0.053 and 0.102, respectively. Both the hybrid TCNet-Fusion and the hybrid ATCNet

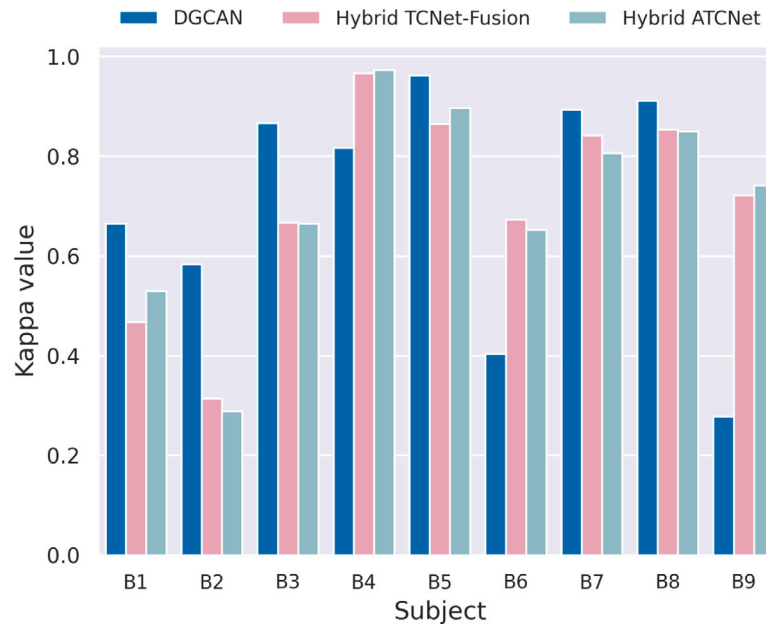


Fig. 15. Kappa score distributions of the top performing hybrid models against the most recent proposal reported in the literature.

Table 8
Performance (% accuracy) comparison of subject-independent classification with state-of-the-art model.

	MSHCNN	EEGNet	Hybrid EEGNet	Deep ConvNet	Hybrid Deep ConvNet	Hybrid TCNet-Fusion	Hybrid ATCNet
B1	76.8	71.28	71.53	71.67	71.66	69.17	70.0
B2	66.32	61.86	65	59.20	59.29	64.56	62.21
B3	57.36	59.29	58.75	58.47	60.05	53.33	58.19
B4	91.75	77.59	84.86	78.91	82.11	84.32	82.70
B5	79.59	79.89	78.78	78.49	76.85	80.81	77.43
B6	82.63	73.34	77.5	71.22	75.65	73.19	68.89
B7	74.16	75.26	77.92	72.13	77.47	76.53	77.36
B8	80.13	73.11	73.68	73.68	73.26	71.97	72.37
B9	75.55	76.33	75.41	76.25	75.99	78.06	76.94
AVG	76.03	72.0	73.72	71.11	72.48	72.44	71.79
P-value		0.060	0.102	0.019*	0.053	0.055	0.052

also displayed no statistical difference when compared to the MSHCNN model.

Fig. 16 visually represents the data distribution of each model. The hybrid Deep ConvNet stands out with the smallest interquartile range, which could indicate consistency and reliability. However, it is essential to mention the presence of outliers in this model and the others, except for the hybrid ATCNet. Remarkably, the Hybrid ATCNet, despite having the lowest accuracy, is the only model that did not exhibit outliers, which contributes to its stability.

It is worth emphasizing that the training of these models was constrained by the slower training times imposed by the quantum simulator. Consequently, we restricted the training duration to 200 epochs without applying any fine-tuning. Despite these limitations, the preliminary results achieved in subject-independent training are on par with state-of-the-art models, underscoring their promise.

For further refinement of subject-independent classification results, consider utilizing the techniques mentioned earlier in conjunction with fine-tuning approaches that incorporate the data of the left out subject in the LOSO approach. These strategies offer the potential to boost model performance. However, it is also essential to explore techniques to mitigate the constraints associated with training in quantum simulators to make this approach more practical.

6. Conclusions

This research introduces two novel BCI systems that leverage quantum computing and machine learning for EEG-based MI classification.

The first approach involves an ML system incorporating a quantum genetic algorithm as a wrapper feature selection method. This approach extracts hand-engineered features and employs a conventional ML classifier. The second BCI system is an end-to-end hybrid classical-quantum architecture that integrates a classical feature extraction network based on baseline CNNs with a quantum classification network based on a dressed quantum circuit. The dressed quantum system comprises a variational quantum circuit with classical preprocessing and postprocessing layers at the beginning and end of the circuit, respectively.

To assess the effectiveness of our proposed methods, we conducted extensive experiments using dataset 2b from the BCI Competition IV. We employed three distinct training scenarios to evaluate the performance of our hybrid models in comparison to various state-of-the-art techniques: subject-dependent cross-validation, subject-dependent hold-out validation, and subject-independent classification.

In the subject-dependent cross-validation approach, we tested both the ML and the end-to-end hybrid systems, with the hybrid models demonstrating superior performance. Notably, the hybrid models achieved competitive results even when using baseline architectures like EEGNet and Deep ConvNet for feature extraction and a general-purpose ansatz for classification. In terms of accuracy, the hybrid EEGNet exhibited no statistically significant difference in subject-dependent cross-validation performance when compared to half of the baseline models, despite inherent limitations. To further emphasize the potential of variational quantum classifiers, we extended our investigations by implementing and testing hybrid models with more complex encoders,

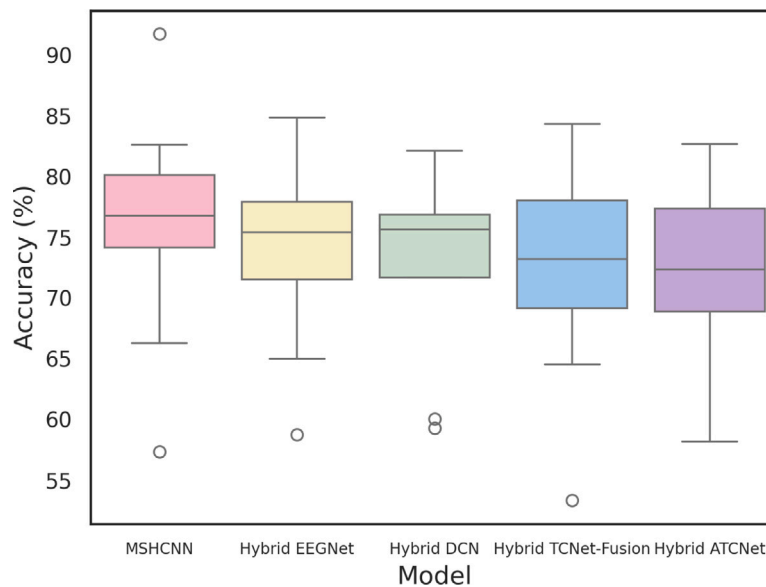


Fig. 16. Box plot of the accuracy (%) scores for each model.

such as ATCNet and TCNet-Fusion, and a shallower ansatz. In the experiments using the subject-dependent hold-out approach, we found that the hybrid ATCNet delivered results akin to state-of-the-art models, exhibiting no statistical differences. As the encoder's performance improved, so did the hybrid network.

Similarly, in the subject-independent setting, the performance of the hybrid models exhibited no statistically significant differences compared to the best-performing baseline model. These compelling outcomes reinforce the validity of our contributions, demonstrating the practical feasibility of Noisy Intermediate-Scale Quantum (NISQ) algorithms in addressing real-world applications. Furthermore, our results highlight the comparable performance of hybrid models with their classical counterparts in both subject-dependent and subject-independent classification, suggesting that the quantum neural network behaves similarly to the fully connected softmax layer in motor imagery classification.

Given the current limitations of quantum technologies, there is no immediate advantage in applying quantum techniques to classify MIEEG signals. However, it is important to note that quantum computers and algorithms are rapidly advancing. The results obtained in this study showcase the promising potential of quantum techniques as promising candidates for accurately classifying complex data, such as EEG signals. We hope these findings will motivate the development and application of dedicated quantum methods specifically designed for EEG signal classification, opening new avenues for future advancements in this field.

In terms of future work, there are several directions that we intend to explore to enhance the proposed approach further. Firstly, we aim to investigate the application of more recent and advanced feature extraction networks to improve the quality of extracted features. Additionally, we plan to explore different ansatz designs for the variational quantum classifier to achieve better classification performance.

Moreover, it is crucial to validate the performance and reliability of the model across diverse datasets. Therefore, conducting experiments on additional datasets will be an essential step in assessing the robustness and generalizability of the algorithm. Moreover, applying various preprocessing techniques can effectively mitigate the challenge of small data sets.

In addition to the enhancements above, we intend to extend the applicability of the hybrid network to address multiclass motor imagery problems. This expansion would involve adapting the network to handle multiple categories of motor imagery tasks, paving the way

for broader practical applications. To accomplish these objectives, we will explore alternative quantum computing models that might offer advantages and novel capabilities tailored to this problem domain.

CRediT authorship contribution statement

Cynthia Olvera: Conceptualization, Investigation, Methodology, Software, Writing – original draft, Writing – review & editing, Validation. **Oscar Montiel Ross:** Conceptualization, Methodology, Formal analysis, Supervision, Validation, Writing – review & editing, Project administration, Funding acquisition. **Yoshio Rubio:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

<https://bbci.de/competition/iv/#download>.

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